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School of Clinical Medicine
Department of Orthopaedics
& Traumatology
香港大學矯形及創傷外科學系

Cervical spine pathology

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Clinical Assistant Professor



Module 1: Imaging of the cervical spine

Cervical X-rays



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- Coverage / adequacy
- Alignment
- Bone
- Disc space
- Soft tissue

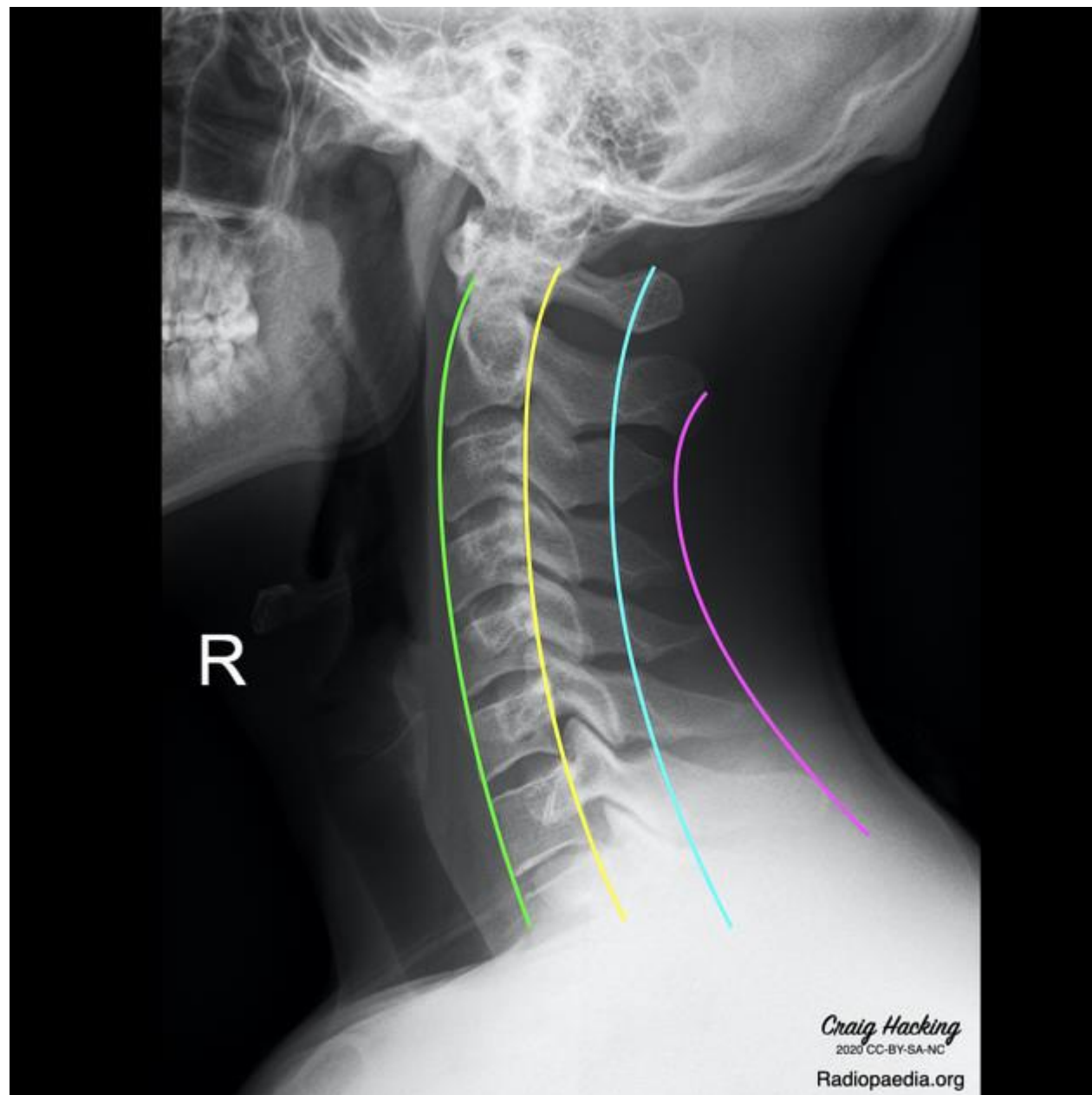
- Dynamic views
- C0/C1/C2 relationship



Adequate coverage should be from the skull base down to to the C7/T1 junction

Alignment

- Anterior vertebral line
- Posterior vertebral line
- Spinolaminar line
- Spinous process tip



Bone and disc height

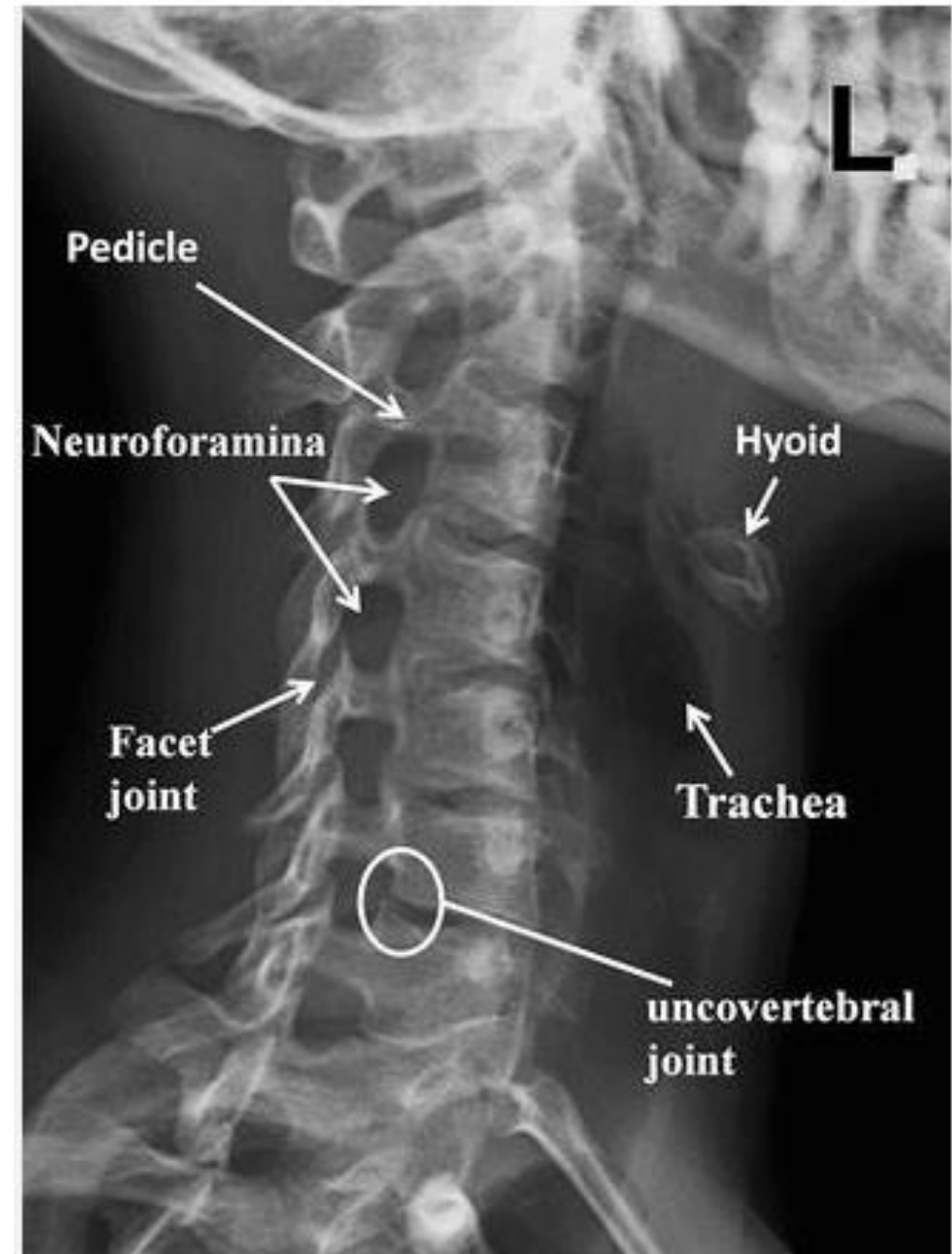
- Destruction of inferior C5 endplate, superior C6 endplate
- Reduced C5/6 disc space
- Resulting in focal kyphosis

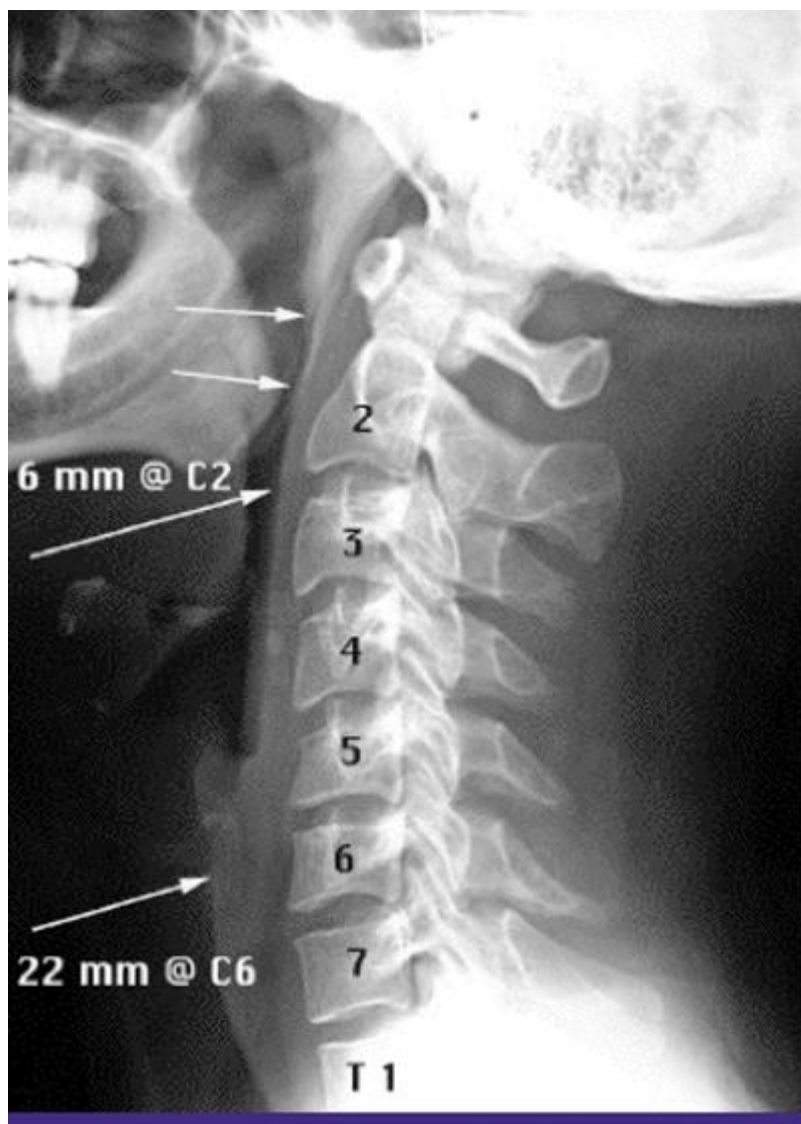
→ Due to bacterial spondylodiscitis affecting the C5/C6 disc



Bone and disc height

- Oblique views facilitate assessment of foraminal narrowing secondary to decrease in disc height and uncovertebral joint hypertrophy





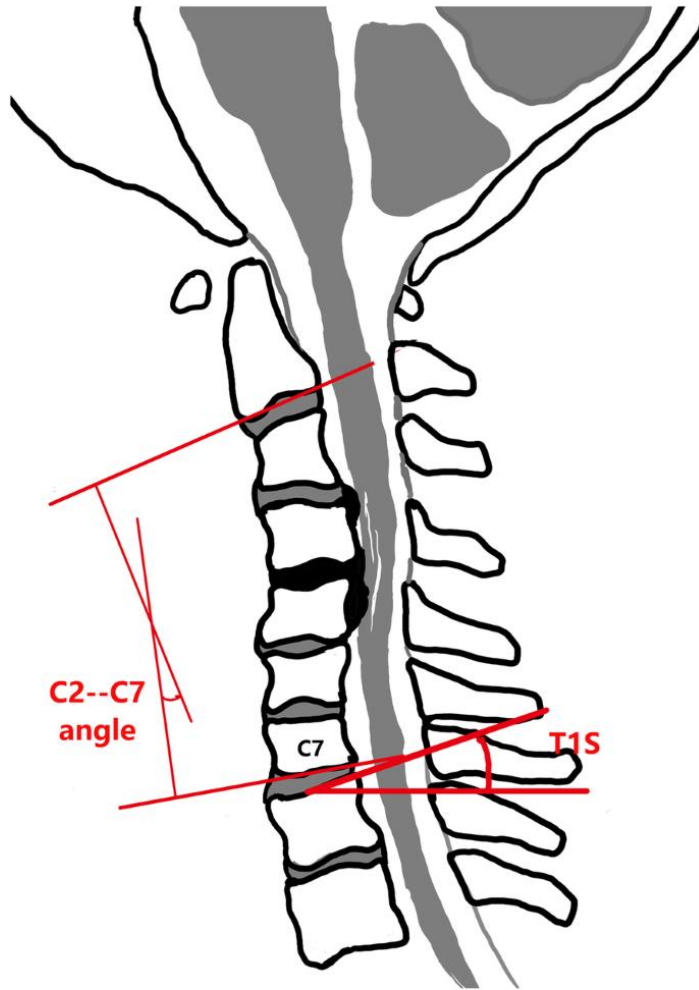
Paravertebral soft tissue

- 6 mm at C2 / $< 1/2$ vertebral depth
- 22 mm at C6 / < 1 vertebral depth



Increased prevertebral soft
tissue swelling

Measurement of cervical alignment

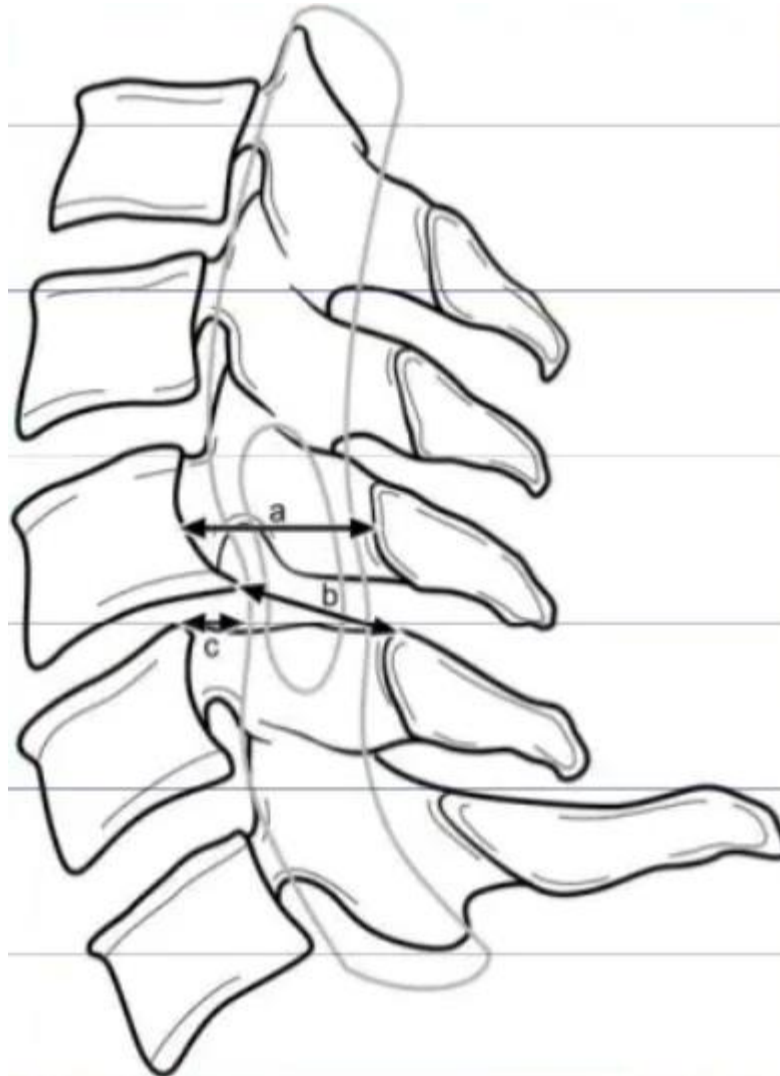


Normal cervical lordosis of 15 – 20 degrees

Measurement of cervical stenosis and instability



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a = Midsagittal diameter of the spinal canal

Relative stenosis if < 12 mm; absolute if < 10 mm

b = Dynamic stenosis = distance from posteroinferior corner of cranial vertebra to anterosuperior edge of caudal lamina

Dynamic stenosis if < 12 mm

c = Olisthesis (slip)

Dynamic instability

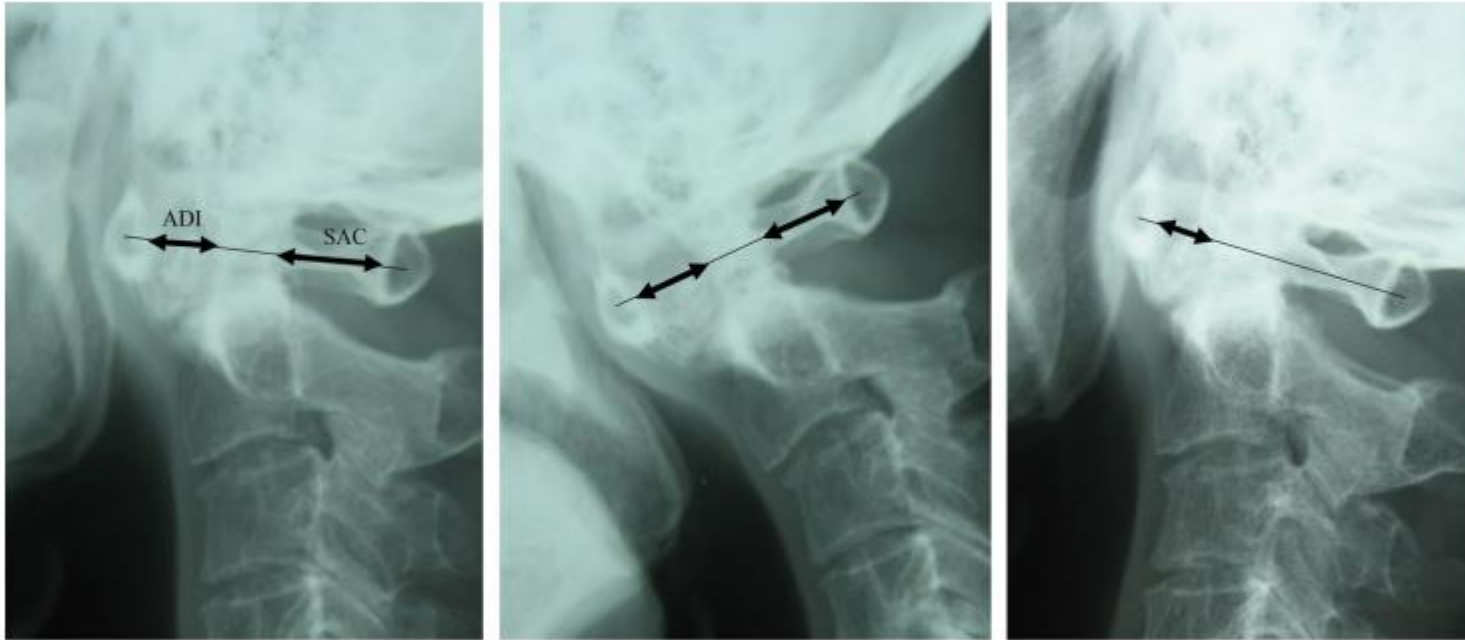


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Dynamic instability if
**positional change >
3.5 mm / > 11°**
between adjacent
vertebra

C1 and C2

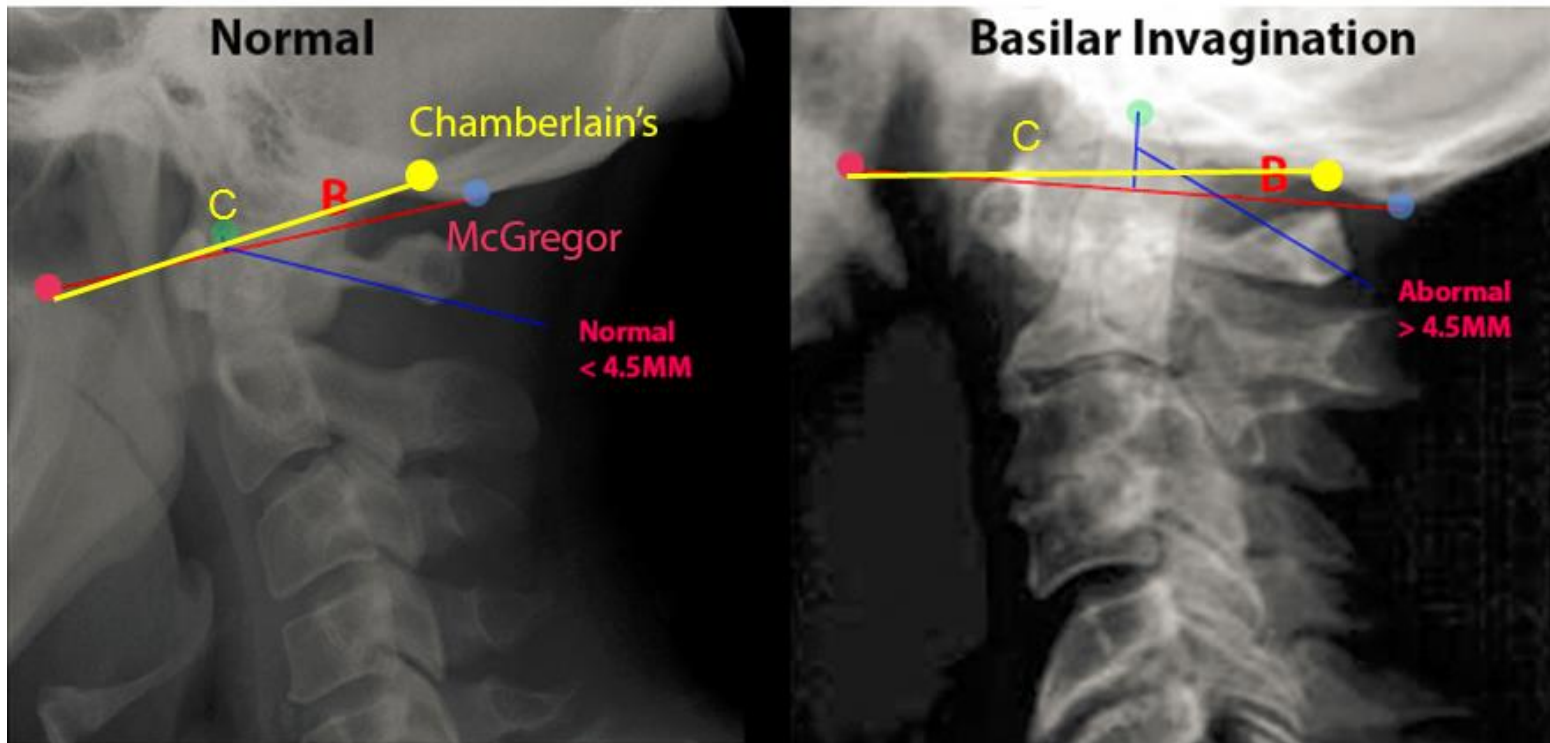


ADI (atlanto-dental interval) should be < 4 mm; assess in flexion and extension views

SAC (space available for the cord) should be > 14 mm

Causes: rheumatoid arthritis, Down's syndrome

Basilar invagination: odontoid prolapsing into the foramen magnum



Chamberlain's line:

Line joining posterior hard palate to opisthion (dens should not pass)

McGregor line:

Line joining posterior hard palate to most caudal point of occipital curve (dens should be < 4.5 mm above)

Causes: congenital (i.e. achondroplasia) and acquired (i.e. rheumatoid arthritis)

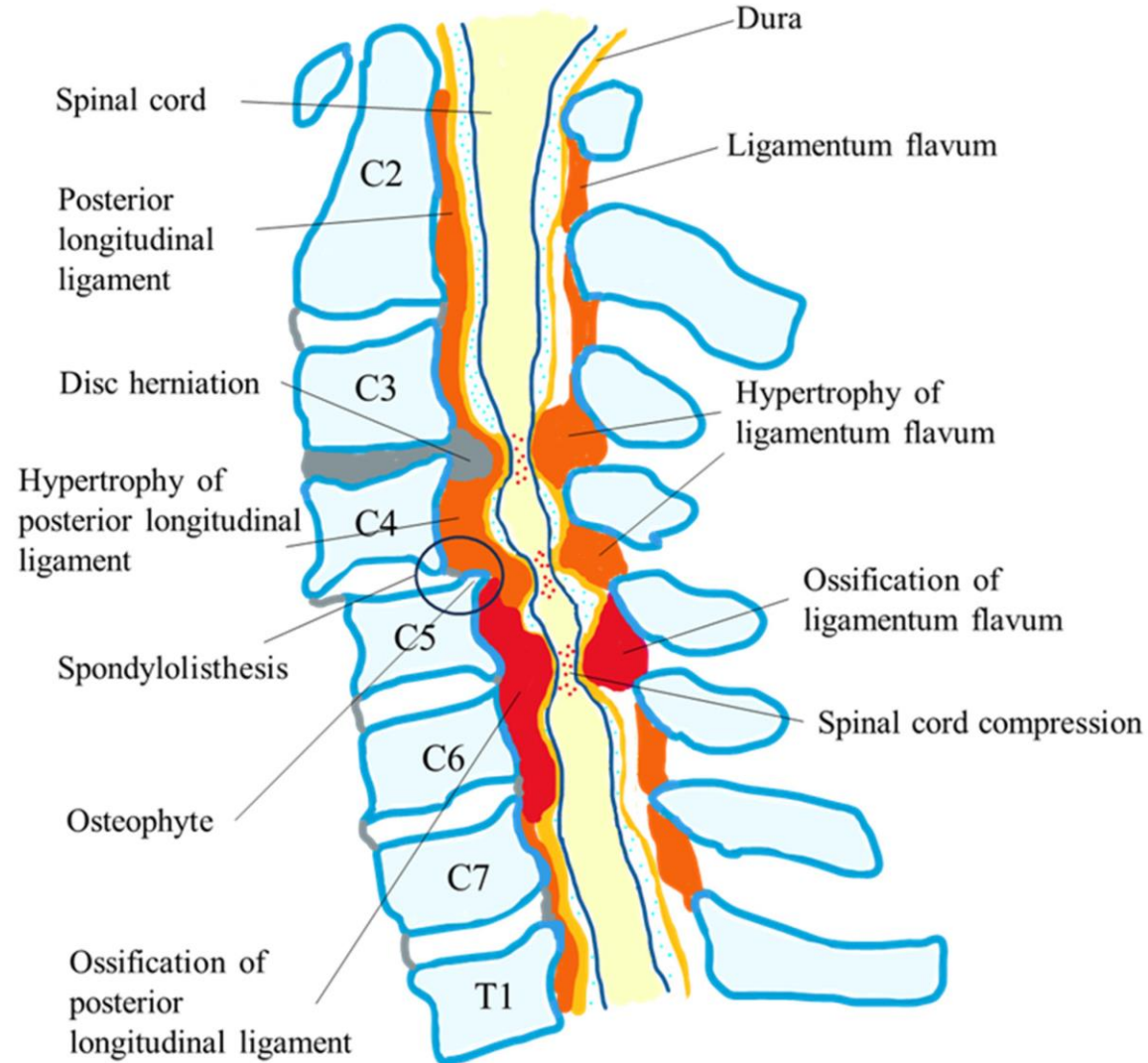


Module 2: Cervical myelopathy and radiculopathy

Pathological causes of canal and foraminal compression



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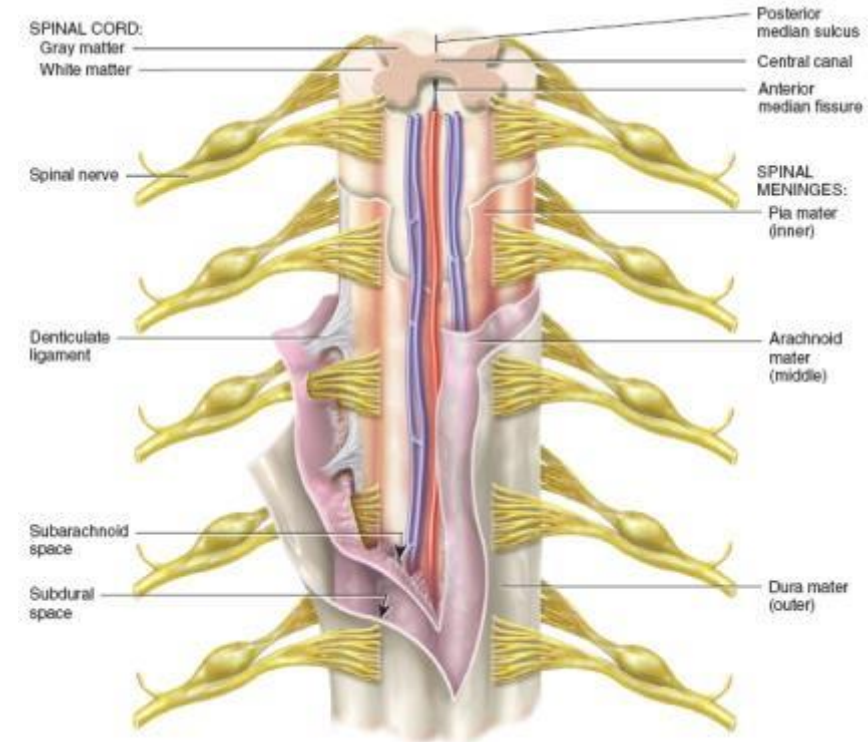


Clinical features of cervical myelopathy



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- Numbness or sensory disturbance (fingers, upper limb)
- Loss of hand dexterity
- Poor proprioception + spastic gait
- Motor weakness and sphincteric dysfunction appear in the late stage
- Pain is **not** a predominant feature
- **Upper motor neuron** signs below level of compression
- **Lower motor neuron** signs at the level of compression



Signs of cervical myelopathy



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- Myelopathic hand signs
 - Hoffman's sign
 - 10 sec test
 - Finger escape sign
- Lower limb spasticity
- Positive Romberg sign / failed tandem walking test
- Lower limb upper motor neuron features

Hoffman's sign



- Involuntary finger flexor contraction due to activation of a monosynaptic stretch reflex
- Positive response represents dis-inhibition of a C8 reflex, indicates lesion above C8



Finger escape sign

- Present the arms with fingers forward and palms down
- Deficient adduction and/or extension of ulnar 2 or 3 fingers



Grade	Fingers	Deficiency
0	All	None
1	Little	Unable to hold adduction
2	Little or little and ring	Unable to assume adduction
3	Little and ring	Unable to assume adduction or full extension
4	Little, ring and middle	Unable to assume adduction or full extension

10 second test



- Number of complete cycles of movement within 10 seconds
- Rhythm of movement
- Difficult, slow and incomplete finger extension
- Exaggerated (failure of synergy between wrist and finger)

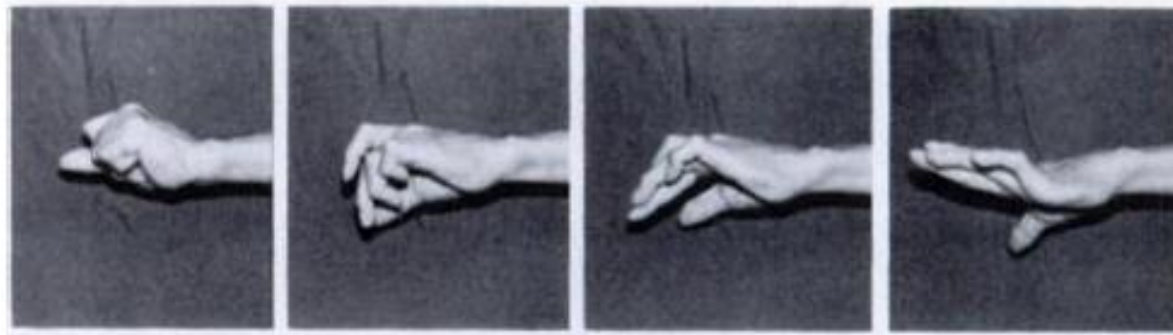


Fig. 3

The grip-and-release test. The number of repeated actions within 10 seconds is recorded. The normal result is 20 or more.

Inverted supinator reflex



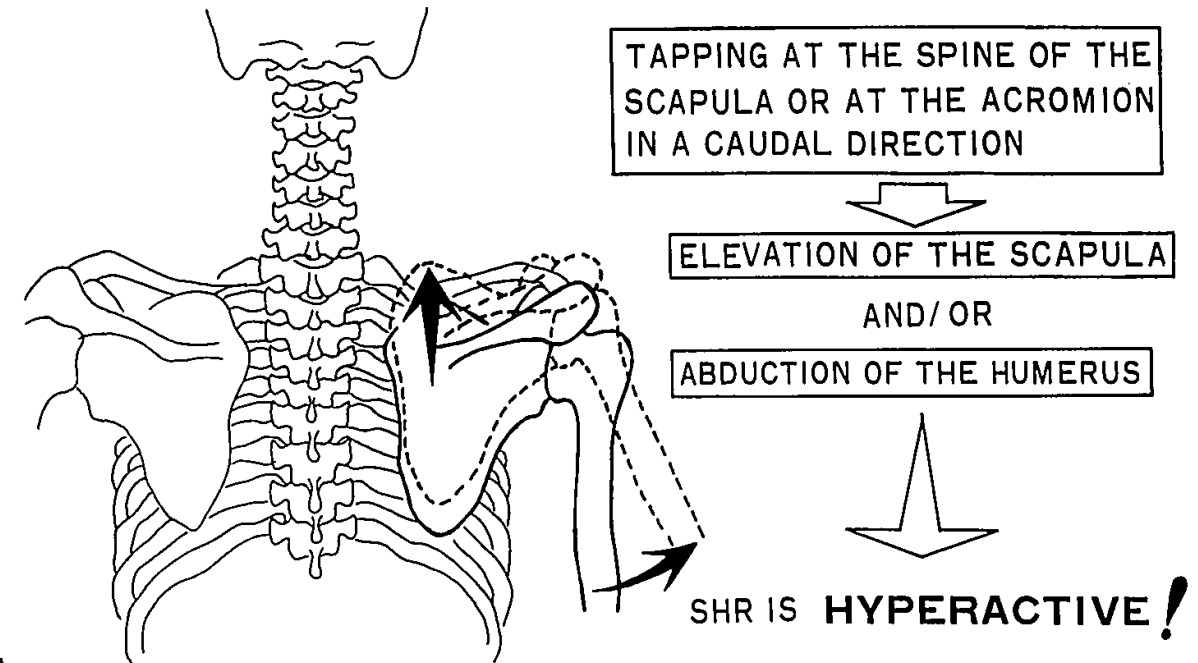
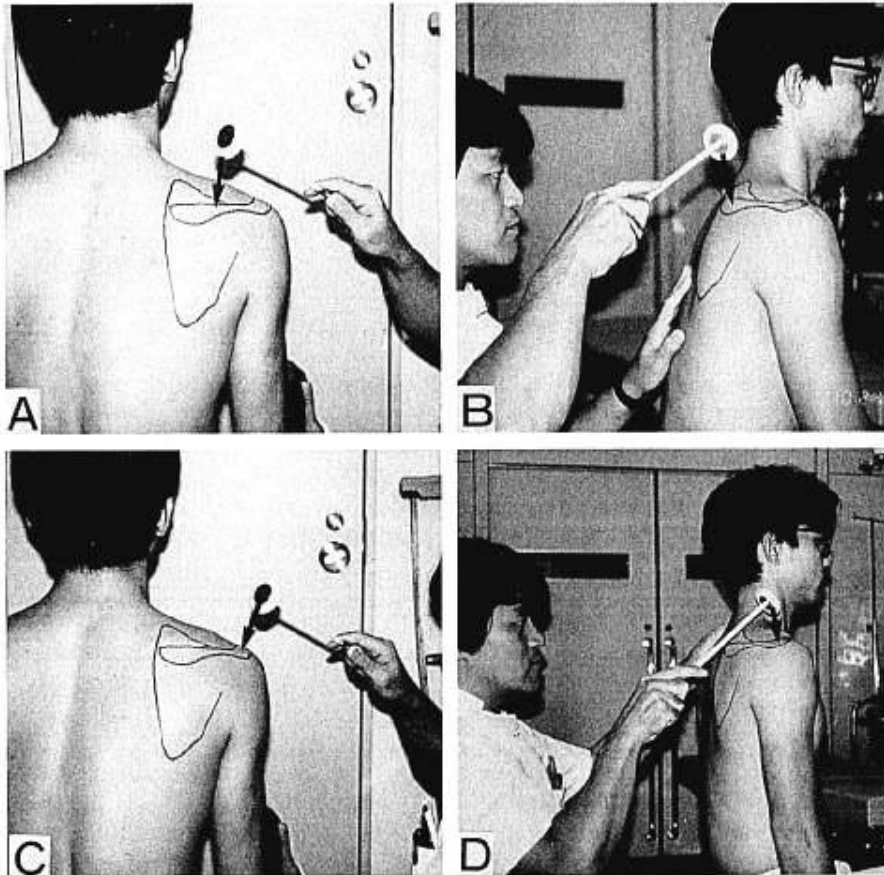
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- Introduced by Babinski in 1910
- Signifying lesion at C5/6 or above
- Two components of abnormal reflex
 - **Absence** of contraction of brachioradialis
 - **Hyperactive** contraction of finger flexors (C8)



Scapulohumeral reflexes

- Scapulohumeral reflex
- Cord compression above C3 vertebral body



Localization of symptomatic disc level



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- Spinal cord ascends approximately one segment during its development
- Cervical discs generally lie opposite spinal cord segments which are numbered one lower than the number of roots passing them

Kokubun et al *Curr Orthop* 1998

Neurologic Level Diagnosis of Cervical Stenotic Myelopathy

Atsushi Seichi, MD, Katsushi Takeshita, MD, Hiroshi Kawaguchi, MD, Ko Matsudaira, MD,
Akiro Higashikawa, MD, Naoshi Ogata, MD, and Kozo Nakamura, MD

Table 2. A New Index of Neurological Level Diagnosis of Cervical Myelopathy

	C3-C4	C4-C5	C5-C6	C6-C7
Reflex	BTR ↑ TTR ↑	BTR ↓ or → TTR ↑	BTR → TTR ↓ or → F.F. ↑	BTR → TTR → F.F. ↓
Uppermost muscle with weakness	Deltoid	Biceps	Triceps or EDC	APB or ADM
Sensory disturbance				

ADM indicates abductor digiti minimi; APB = abductor pollicis brevis; BTR = deep tendon reflexes of biceps; EDC = extensor digiti communis; FF = finger flexor reflex; TTR = deep tendon reflexes of triceps.

Other assessments



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Long tract signs

- Lower limb hyper-reflexia
- Clonus and upgoing Babinski
- Dorsal column function
 - Romberg
 - Tandem walking
 - Lhermitte sign (severe demyelination and hyper-excitability of dorsal column)

Sphincter function

- Urinary frequency
- Retention
- Overflow incontinence
- Change in bowel habit

Functional assessment

- Japanese Orthopaedic Association score

Modified JOA score



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- Four scoring categories
 - Upper limb motor and sensory
 - Lower limb function
 - Urinary sphincter
- Score from 0 - 17

Guide of disease severity, facilitates decision making for surgery, and monitoring of disease progression / recovery

Advanced imaging assessment



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- Evaluation of compression and deformity of spinal cord
- Evaluation of intramedullary lesion
- Detection of pathological spinal factors
- Surgical planning

MRI – basic interpretation

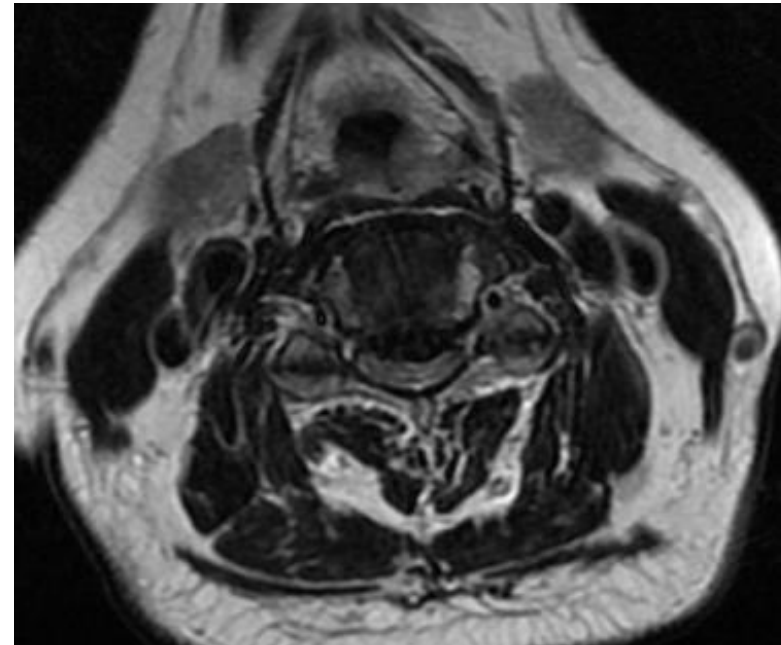
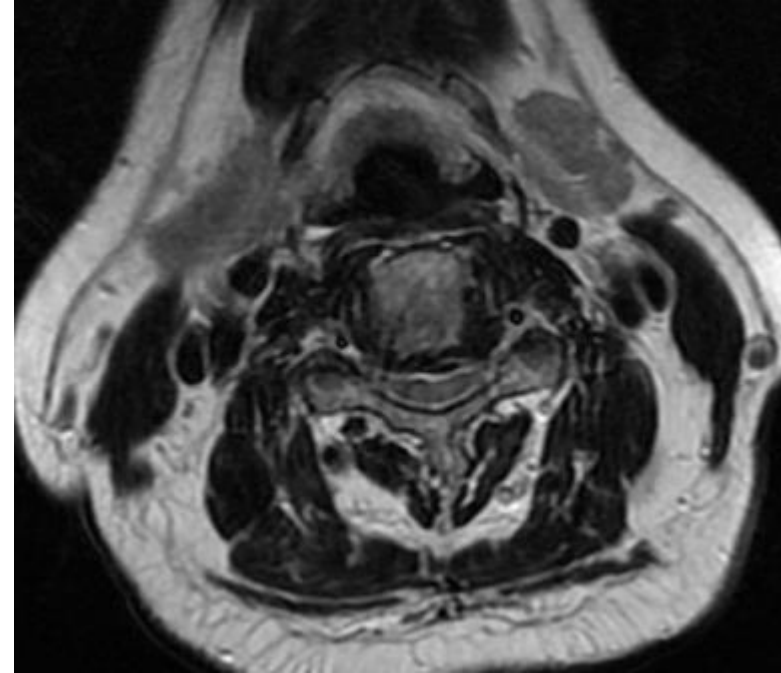
- Level of lesion, location
- Pathoanatomy (disc, osteophyte, OPLL, flavum)
- Obliteration of the CSF space
- Cord shape / cross-sectional area
- Intramedullary signal change (myelomalacia)



Cord compression at C3/4 with myelomalacia



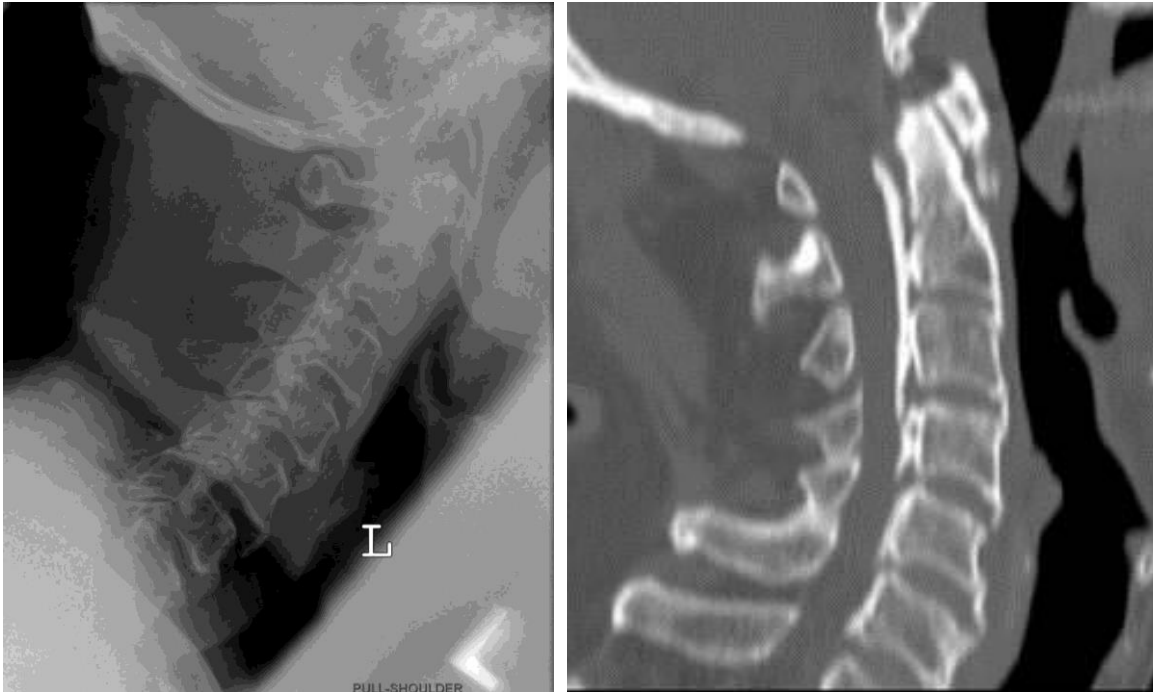
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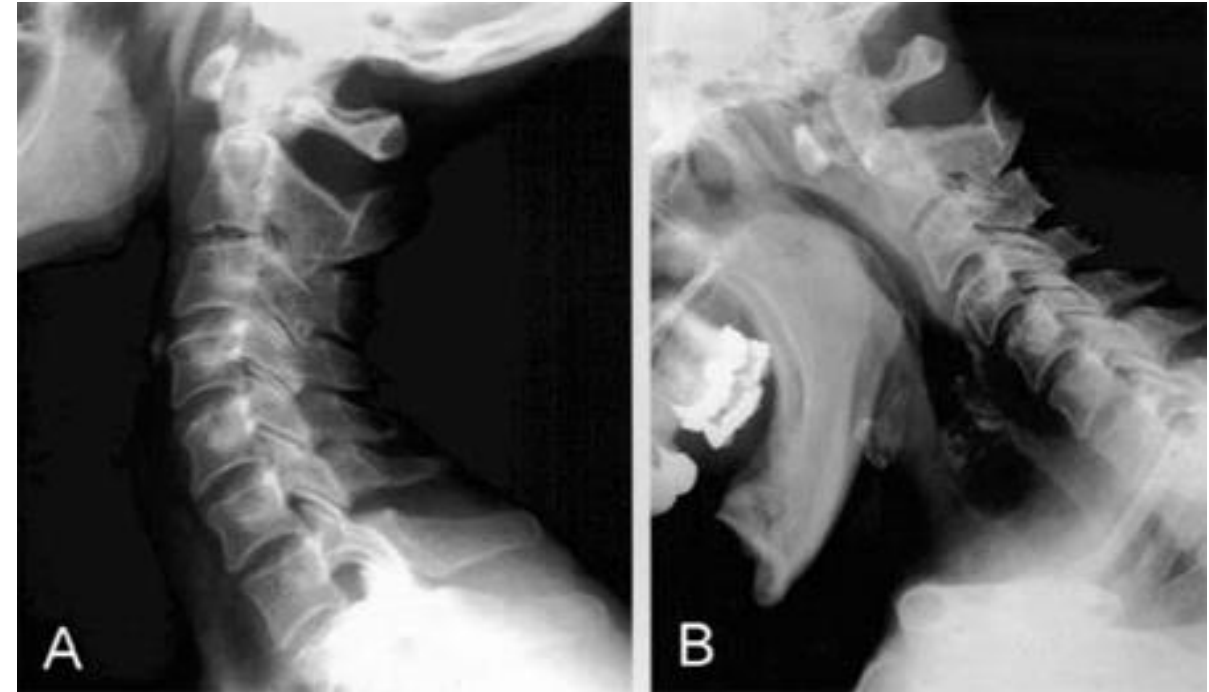
Important to rule out other pathological process



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Plain CT scan to delineate ossification of posterior longitudinal ligament



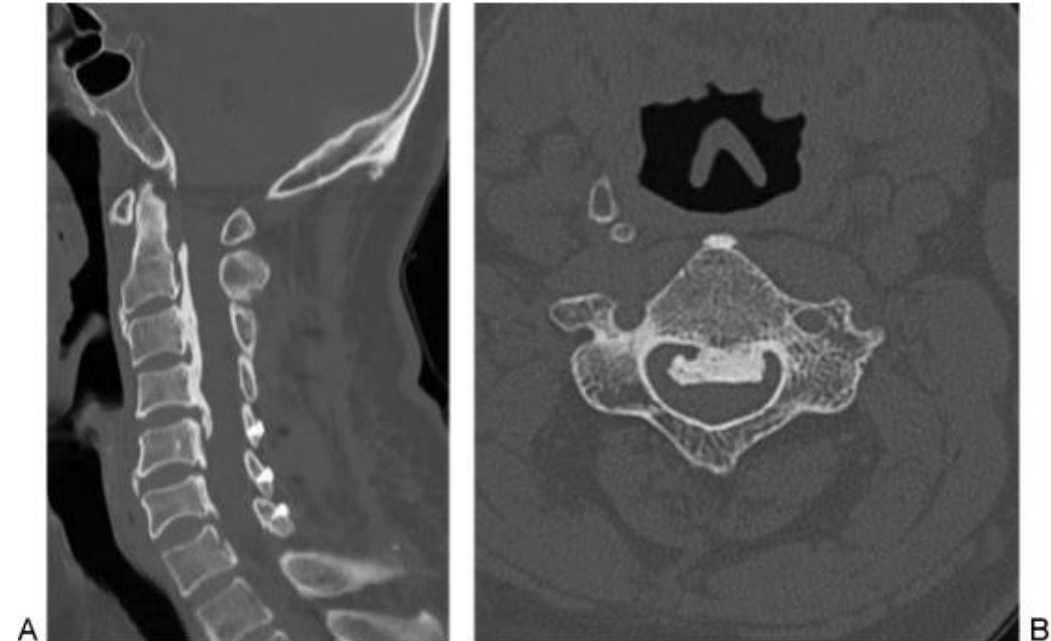
Dynamic X-rays to diagnose atlantoaxial subluxation in rheumatoid arthritis

OPLL – Ossification of the posterior longitudinal ligament



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- Ectopic calcification as a cause of cord compression
- Much more common in Asian vs non-Asian populations (up to 2-4%)
- Male predominance, affects younger population
- More commonly affects cervical than thoracic spine

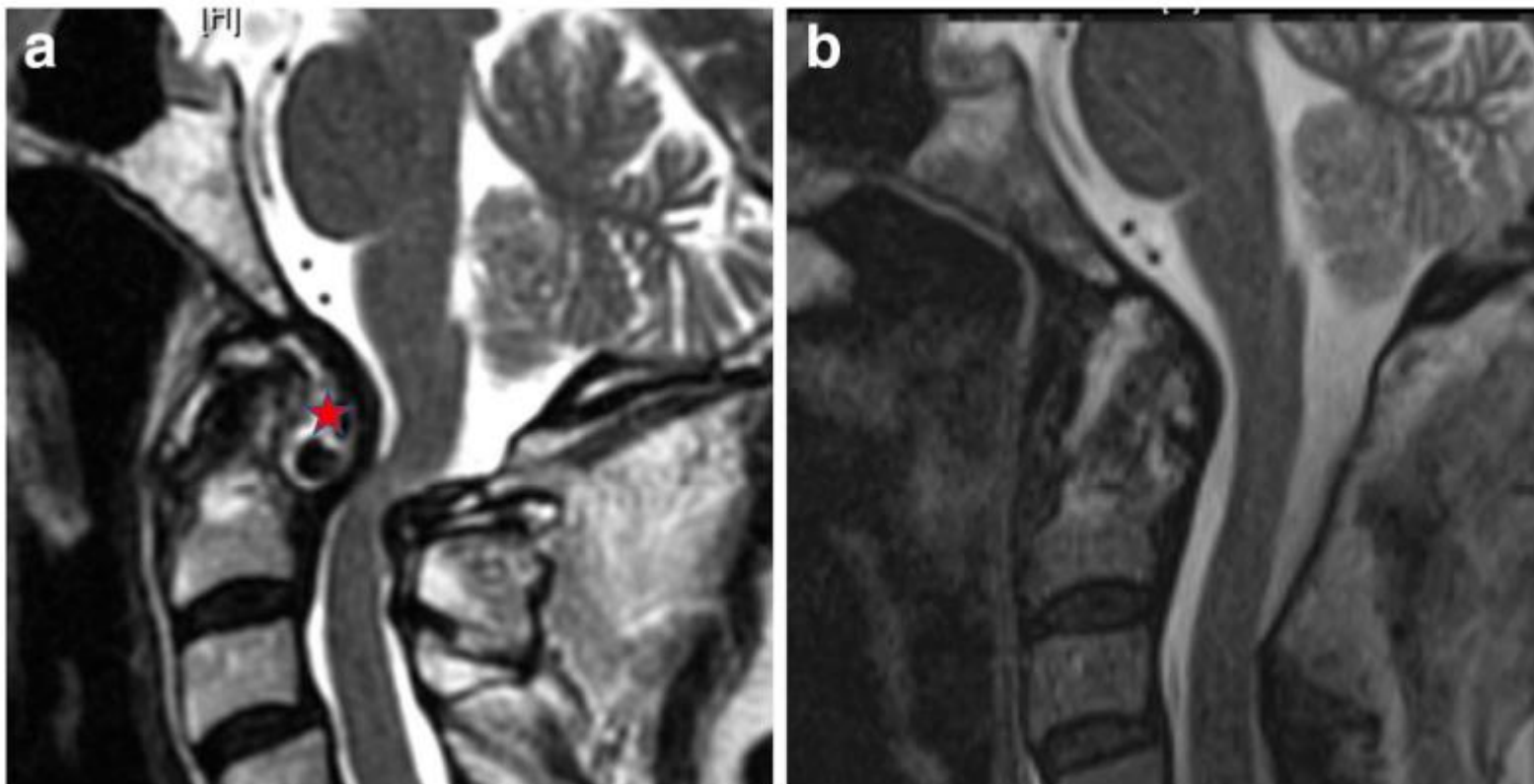


Rheumatoid arthritis and the cervical spine



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- High prevalence of cervical spine involvement
- Formation of retroodontoid pannus
- Synovial inflammation predisposes to cervical instability
 - Atlantoaxial subluxation (C1/2)
 - Subaxial subluxation
 - Vertical subluxation (basilar invagination)
- Use of RA medications reduces the incidence of cervical involvement



Left - retroodontoid mass causing cervical spinal cord compression
Right – after posterior decompression

Natural history of CSM



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- Progressively deteriorating disorder
 - 75% deteriorates in a **stepwise** fashion
 - 20% deteriorates **slowly and steadily**
 - 5% has **rapid onset** with a **stable plateau**
- At present, no good conservative treatment options

CERVICAL MYELOPATHY: A COMPLICATION OF CERVICAL SPONDYLOSIS

BY

EDWIN CLARKE AND PETER K. ROBINSON

*The Postgraduate Medical School of London, and the
National Hospital, Queen Square*

With a Surgical Note by L. S. Walsh and Ian Mackenzie

"I am inclined by long observations of spinal affections, to think that the ligaments are more frequently in fault than any of the structures of the spinal column; and that many of the cases of paralysis, whether temporary or permanent, and vaguely classed under functional derangements of the spinal cord, will be found, by dissection, to depend on an altered condition of those textures."

C. ASTON KEY, 1838.

THE effects of acute injury of the cervical spinal cord were known to the Ancient Egyptians (Breasted, 1930, Cases 29 to 33, 48). That chronic trauma due to bony and ligamentous changes also occurs is, despite the foresight of Aston Key (1838), a recent concept and we propose to review this condition in the light of the literature and the experience of 120 cases.

The term "cervical spondylosis" (Schmorl and Junghanns, 1932; Forsberg, 1939) refers to chronic degenerative changes due primarily to intervertebral disc decay (Bailey and Casamajor, 1911; Beadle, 1931; Keyes and Compere, 1932; Horwitz, 1940; Saunders and Inman, 1940; Brain, 1954a), which are probably universally present in elderly persons. In many this process is symptomless but occasionally neurological complications—root or cord compression—result. If cord involvement occurs, the clinical picture of cervical spondylotic myelopathy is produced. The term "cervical disc" is usually applied to this condition but as it does not describe the cord lesion and as the disc is only partially responsible for the spondylosis, it should be used no longer. The true incidence of cervical myelopathy is unknown but as Pallis, Jones and Spillane (1954) have shown, mild forms, at times manifest only objectively, may be relatively common.

Cord involvement may also occur as a result of an acute prolapse of the cervical intervertebral disc and it is of great importance to differentiate this from the more chronic cervical myelopathy, especially as the pathology (Brain, 1948; Frykholm, 1951), prognosis and treatment are different. It is

Indications for surgery in CSM



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- Progressive neurologic deficit
- Significantly impaired ADL (JOA score)
- Compatible imaging findings



General considerations:

Anterior vs posterior approach

1. Sagittal alignment

- Fixed kyphotic deformities favours anterior approach

2. Pathoanatomy

- Disc protrusions into the spinal cord
 - better treated with anterior technique
- Infolding / thickening of ligamentum flavum
 - better treated with posterior technique

3. Number of levels involved

- 3 level pathology or more favours posterior approach

General considerations:

Anterior vs posterior approach



4. Subluxation or instability

- Fusion is indicated

5. Neck pain

- Laminoplasty patients may experience more post-op axial neck pain

Examples:



Laminoplasty



1 level ACDF

Counseling patients on complications



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- **Anterior surgery:**

- esophageal injury (intra-op or late due to implant)
- vertebral artery injury
- airway compromise
- Pseudoarthrosis (failed fusion)
- adjacent level degeneration

- **Posterior surgery:**

- post-op kyphosis
- axial symptoms (shoulder and neck pain)
- C5 nerve root paresis

Clinical features of cervical radiculopathy



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- Unilateral arm pain or sensory disturbance
- May have associated weakness
- Dermatomal distribution
- Neck pain

Signs of cervical radiculopathy



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- Spurling's test
- Shoulder abduction test
- Myotomal weakness
- Reduced upper limb reflexes



Level	Muscle weakness	Sensory deficits/location of pain	Reflex
C5	Deltoid	Lateral arm	Biceps
C6	Biceps, wrist extension	Radial forearm, radial two digits	Brachioradialis
C7	Triceps, wrist flexion	Middle finger	Triceps
C8	Finger flexors	Ulnar two digits	
T1	Hand intrinsic	Ulnar forearm	

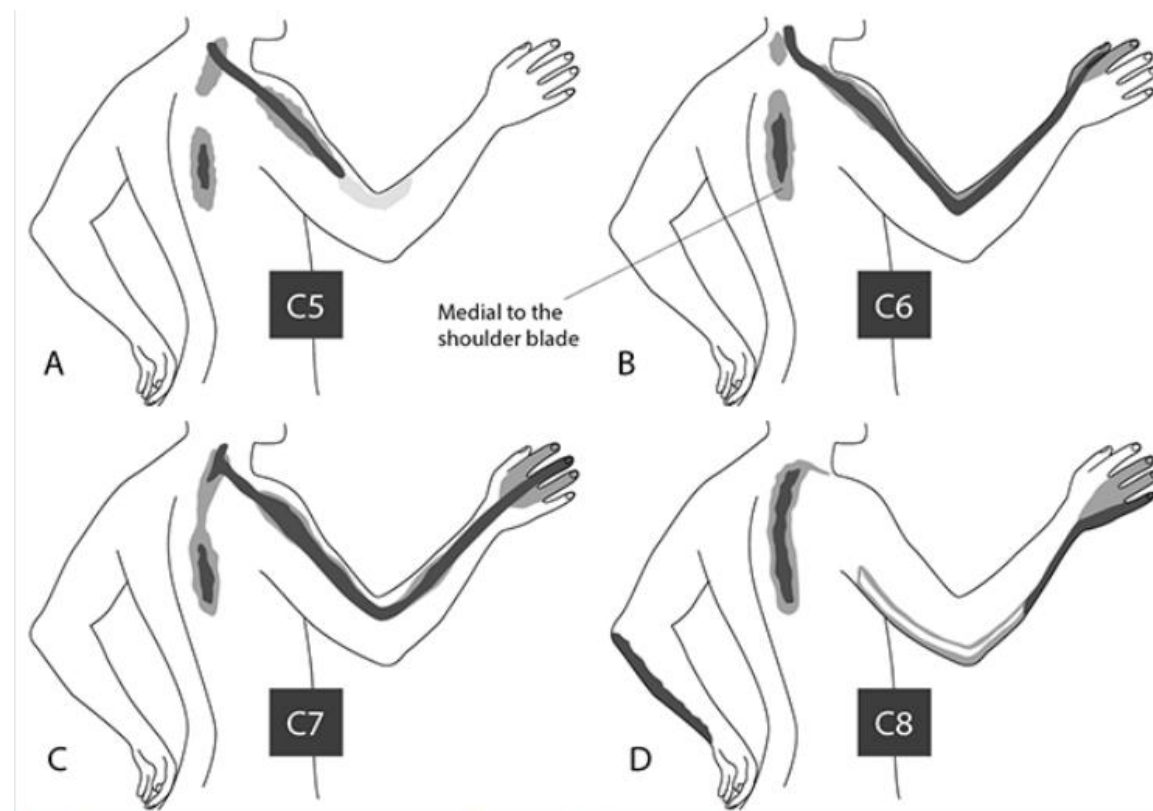


Fig. 1. Pain patterns in cervical radiculopathy C5 - C8.

Differential diagnosis



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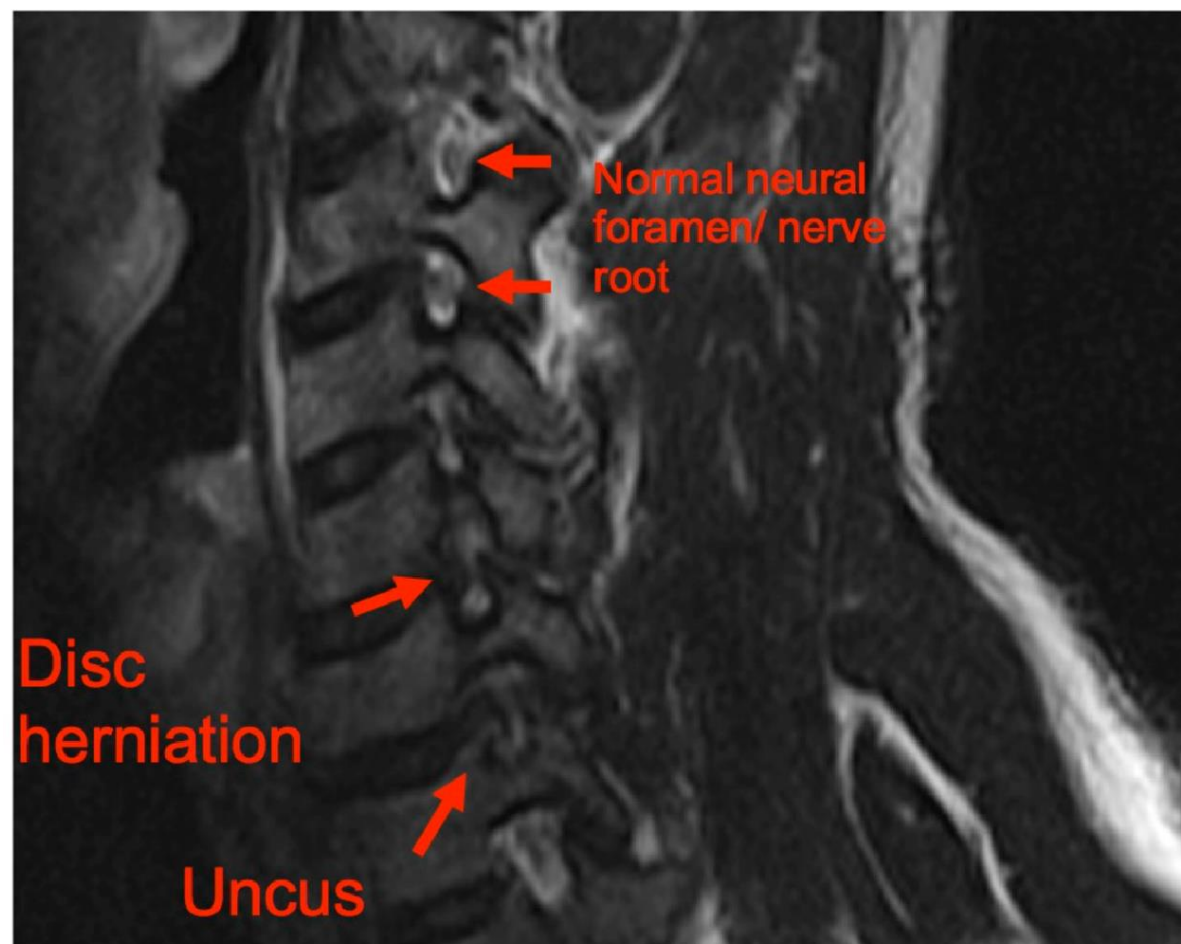
- Cervical myelopathy; symptoms may overlap and patients can have myeloradiculopathy
- Peripheral nerve compression
- Shoulder pathology

Investigations



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- Xrays
 - AP / lateral
 - Oblique views for foraminal narrowing
- CT scan
- MRI scan
 - Confirm nerve root / cord compression
- NCV / EMG



Non-operative treatment of cervical radiculopathy



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- Limited evidence for use of neck collar and bedrest
- Physical therapy
 - Traction
 - ROM exercises
 - Strength training and stretching

Non-operative treatment of cervical radiculopathy



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- Medications
 - NSAIDs
 - Neuroleptics
 - Steroids
- Epidural steroid injections

Up to 90% of patients have good response to conservative treatment measures

Surgical treatment

- Indications
 - Persistent symptoms despite conservative mx
 - Neurological deficits
- Treatment options
 - Anterior: anterior cervical discectomy and fusion, artificial disk replacement
 - Posterior: posterior cervical foraminotomy



Module 3: Cervical trauma and spinal cord injury

Case presentation



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- 21 yo male
- Dived into shallow end of the swimming pool headfirst
- Sudden onset of neck pain
- Unable to move immediately
- Rescued from the pool

What is wrong with this XR?



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1. Nothing
2. Increased soft tissue thickness
3. Malalignment
4. Fracture
5. Inadequate XR – can't tell





Overview



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- Epidemiology
- Emergency management
- Initial assessment
- Complete vs Incomplete lesions
- Spinal cord injury vs Spinal fractures

Epidemiology of spinal cord injury



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– Bimodal distribution

- Young adults
 - High energy trauma
- Old adults
 - Low energy trauma
 - Osteoporotic bone
 - Pre-existing spinal stenosis

Polytrauma and spinal injury



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- Polytrauma involves injuries to the spine in up to 30% of cases
- Multiple spinal fractures occur in 3 to 5%
 - Examine and XR whole spine

Transfer



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- From scene of accident to hospital
- From A&E to ward
- Continue immobilization until clinical and radiological clearance
 - Cervical spine immobilization
 - Spine board
 - Log roll

Emergency Management



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- ATLS for assessment and resuscitation
 - Airway maintenance and cervical spine protection
 - Breathing
 - Circulation
 - Disability – motor / sensory exam
 - Exposure – includes log-roll to inspect and examine spine and per-rectal exam

Emergency Management



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- Trauma series
 - CXR
 - AP pelvis
 - Lateral cervical spine
- Polytrauma CT

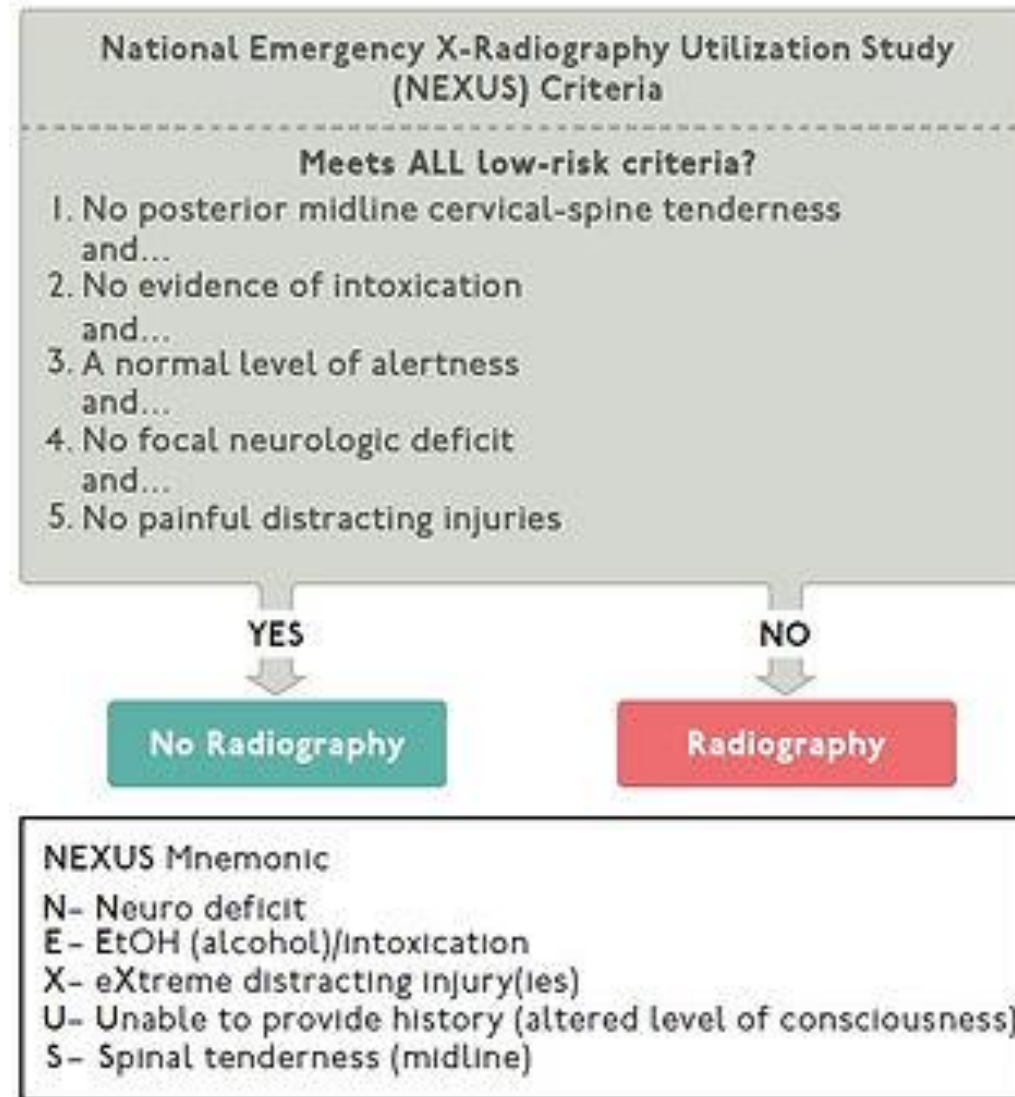
Other radiographic investigations



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- CT scan
 - Readily available, provides radiographic clearance when X-rays are inadequate
- MRI scan
 - Essential if there are neurological deficits
 - Useful for delineation of soft tissue injury (i.e. discoligamentous)

NEXUS criteria to clear the cervical spine



Cervical Spine Stabilization

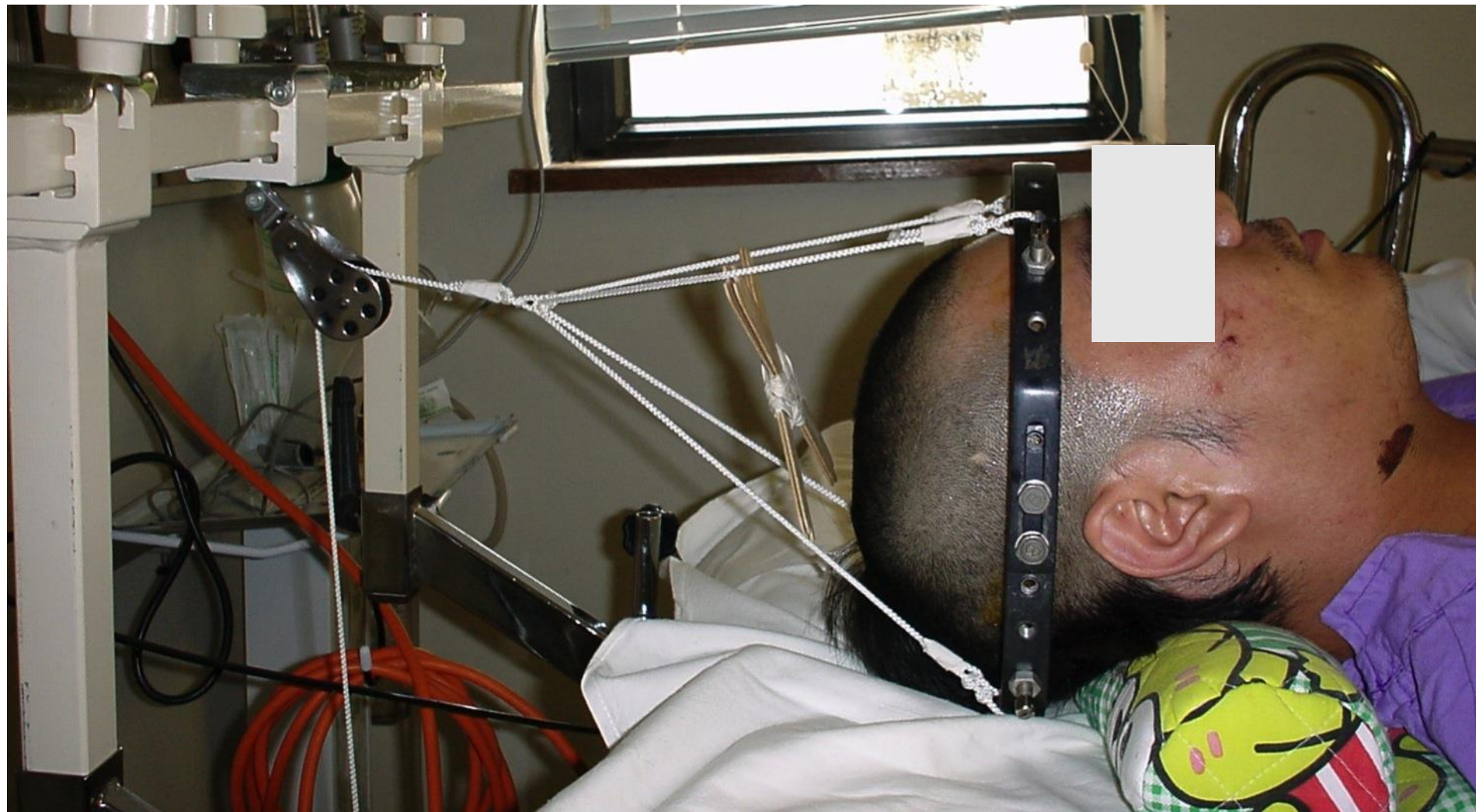


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- Rigid collar (correct size)
- Sand-bags and tape
- Skull traction



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Patient evaluation

- History
 - High / low energy trauma
 - Motor vehicle accident
 - Fall from height
 - Fall from level ground
 - Time of injury
 - Delay in diagnosis
 - Unconscious
 - Head injury
 - Alcohol
 - Multiple distracting injuries

Physical Examination



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- Head to Toe
 - Head / chest wounds and cervical injury
 - Relates to mechanism of injury
- Spine
 - Bruising
 - Tenderness
 - Stepping
 - Restricted movement

Neurological exam

- Aim
 - Assess level of injury/lesion
 - Assess severity of neurological damage
 - Need for intervention
 - Prognosis

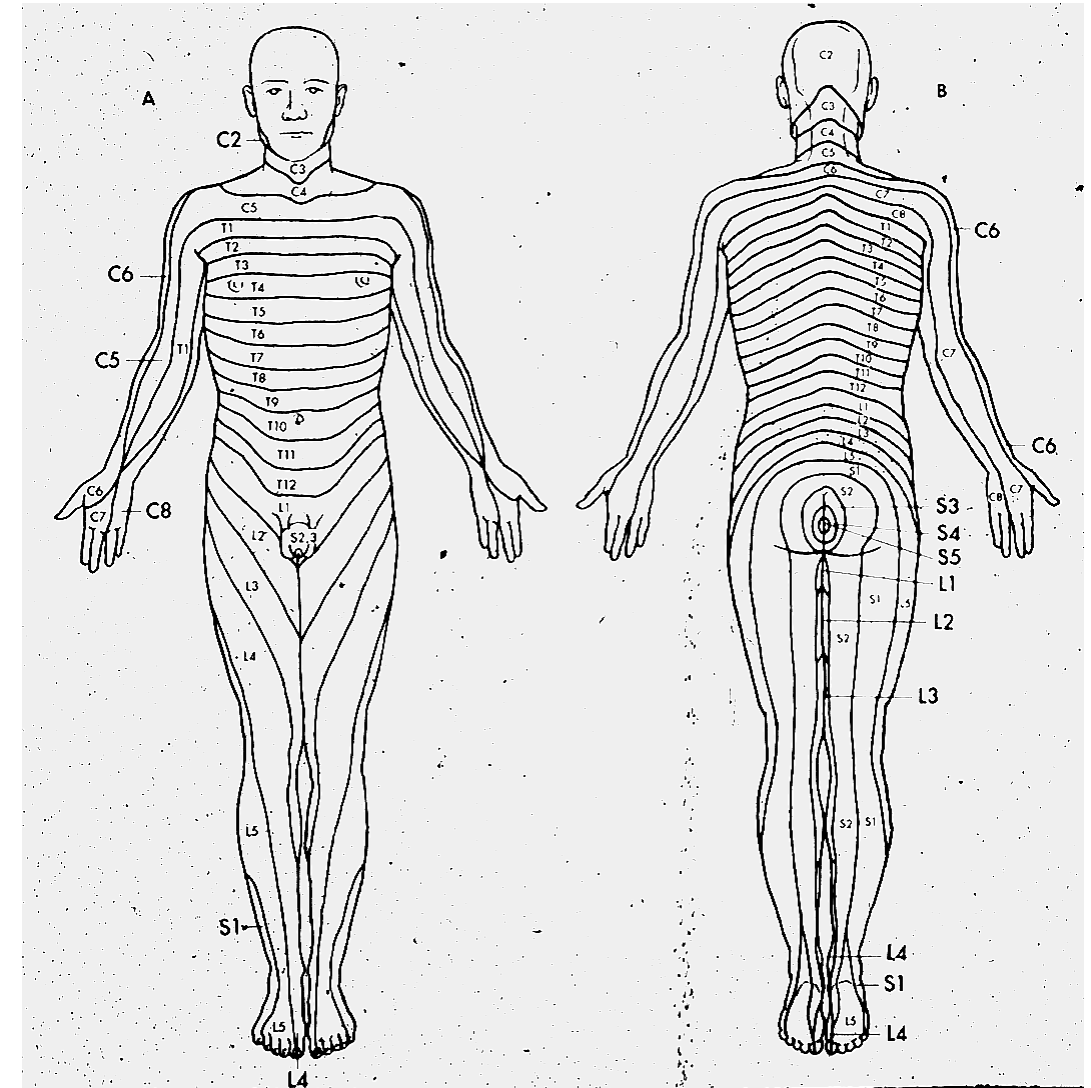
Neurological Evaluation

- Sensory
- Motor
- Reflexes / Sphincter

Sensory

– Dermatome landmarks

Nipple	T4
Xiphoid	T7
Umbilicus	T10
Groin	L1
Anus	S2,3,4



C5 *Elbow flexors*

C6 *Wrist extensors*

C7 *Elbow extensors*

C8 *Finger flexors*

T1 *Finger abductors (little finger)*

L2 *Hip flexors*

L3 *Knee extensors*

L4 *Ankle dorsiflexors*

L5 *Long toe extensors*

S1 *Ankle plantar flexors*

0 = total paralysis

1 = palpable or visible contraction

2 = active movement, gravity eliminated

3 = active movement, against gravity

4 = active movement, against some resistance

5 = active movement, against full resistance

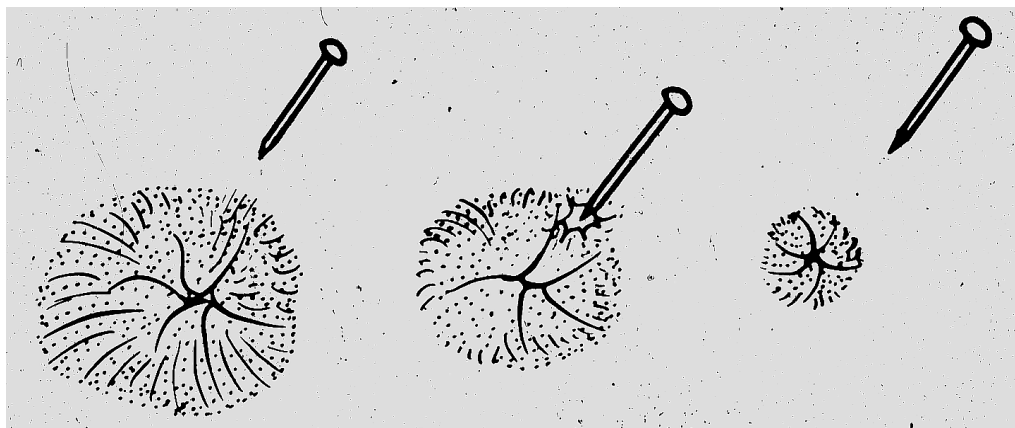
} Full ROM

Spinal shock

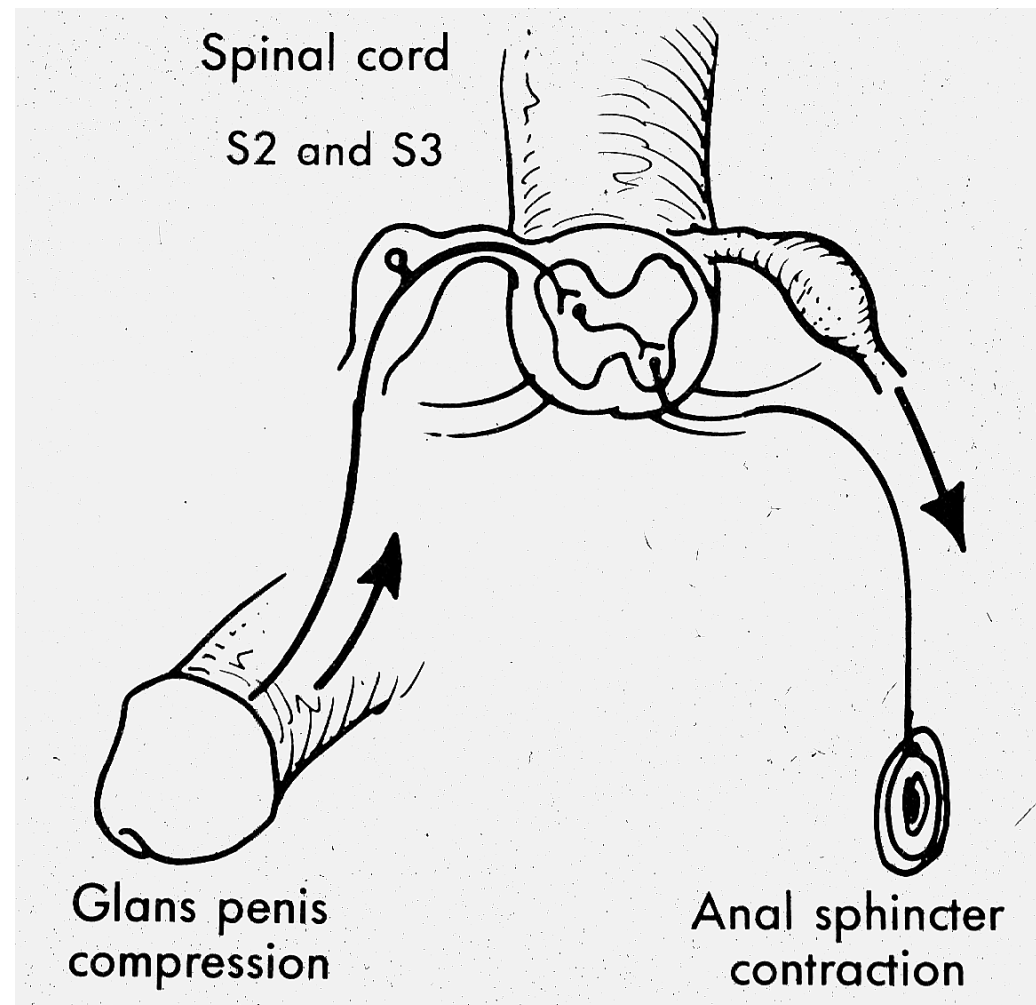


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- Period of temporary loss of function after injury (hours to days)
 - Flaccid paralysis below level of lesion
 - Areflexia
 - Loss of bowel / bladder control
- Spinal shock is over if bulbocavernous and/or anal wink reflex has returned
- *Assess for motor and sensory function only after spinal shock is over*



Anal wink



Bulbocavernosus reflex

Patient Name _____

Examiner Name _____ Date/Time of Exam _____



STANDARD NEUROLOGICAL CLASSIFICATION OF SPINAL CORD INJURY



MOTOR

KEY MUSCLES
(scoring on reverse side)

	R	L	
C5	☐	☐	Elbow flexors
C6	☐	☐	Wrist extensors
C7	☐	☐	Elbow extensors
C8	☐	☐	Finger flexors (distal phalanx of middle finger)
T1	☐	☐	Finger abductors (little finger)

UPPER LIMB TOTAL (MAXIMUM) ☐ + ☐ = ☐ (25) (25) (50)

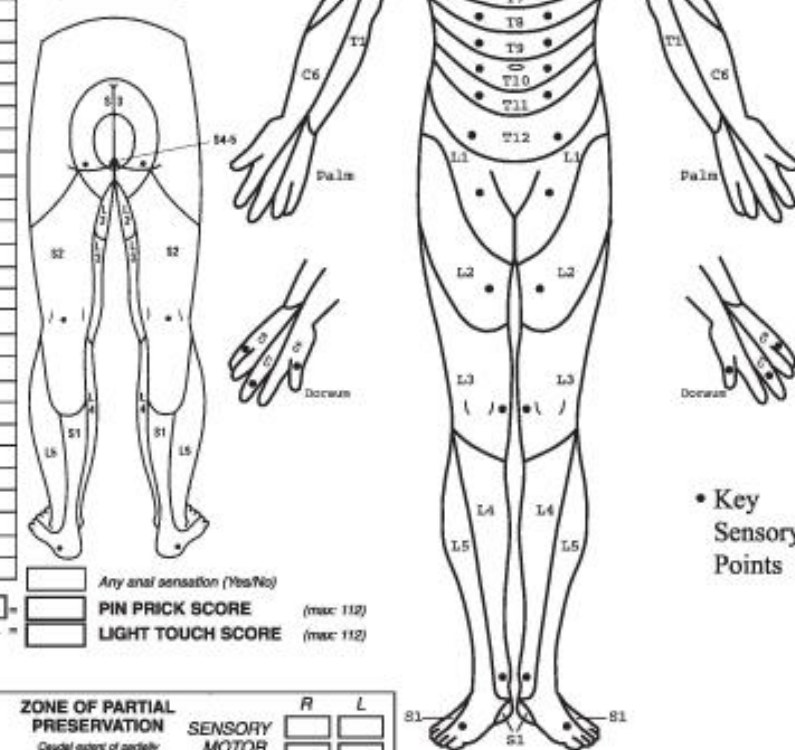
SENSORY

KEY SENSORY POINTS

0 = absent
1 = impaired
2 = normal
NT = not testable

	LIGHT TOUCH		PIN PRICK	
	R	L	R	L
C2				
C3				
C4				
C5				
C6				
C7				
C8				
T1				
T2				
T3				
T4				
T5				
T6				
T7				
T8				
T9				
T10				
T11				
T12				
L1				
L2				
L3				
L4				
L5				
S1				
S2				
S3				
S4-5				

TOTALS: ☐ + ☐ = ☐ (MAXIMUM) (56) (56) (56) (56)



• Key Sensory Points

Comments: _____

Voluntary anal contraction (Yes/No) ☐

Any anal sensation (Yes/No) ☐

LOWER LIMB TOTAL (MAXIMUM) ☐ + ☐ = ☐ (25) (25) (50)

NEUROLOGICAL LEVEL: The most caudal segment with normal function

COMPLETE OR INCOMPLETE? ☐ Incomplete = Any sensory or motor function in S4-S5

ZONE OF PARTIAL PRESERVATION: Caudal extent of partially innervated segments

ASIA IMPAIRMENT SCALE: ☐

Motor level: Lowest myotome with a grade of at least 3 on manual muscle testing

Sensory level: most caudal normally innervated dermatome for both pin prick and light touch sensation

Neurological level of injury: most caudal segment of the spinal cord with normal sensory and antigravity motor function (MMT 3 or more) on both sides

Prognosis



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- Complete injury - poor
 - no motor or sensory recovery, including sacral roots, distal to the site of injury and at the **end of spinal shock**
- Incomplete injury - good
 - Degree of sparing
 - Speed of recovery

Examples of unstable injuries with neurological compromise



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- Bilateral facet dislocation
 - Flexion-distraction injury
 - Bony and ligamentous instability
 - Associated with significant neurological deficit
 - Require emergent treatment via reduction, decompression, and surgical fusion



Examples of unstable injuries with neurological compromise



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- Flexion teardrop fracture
 - Results from flexion and axial loading
 - Triangular bony fragment over anteroinferior body
 - +/- Posterior displacement of vertebral body with canal compromise

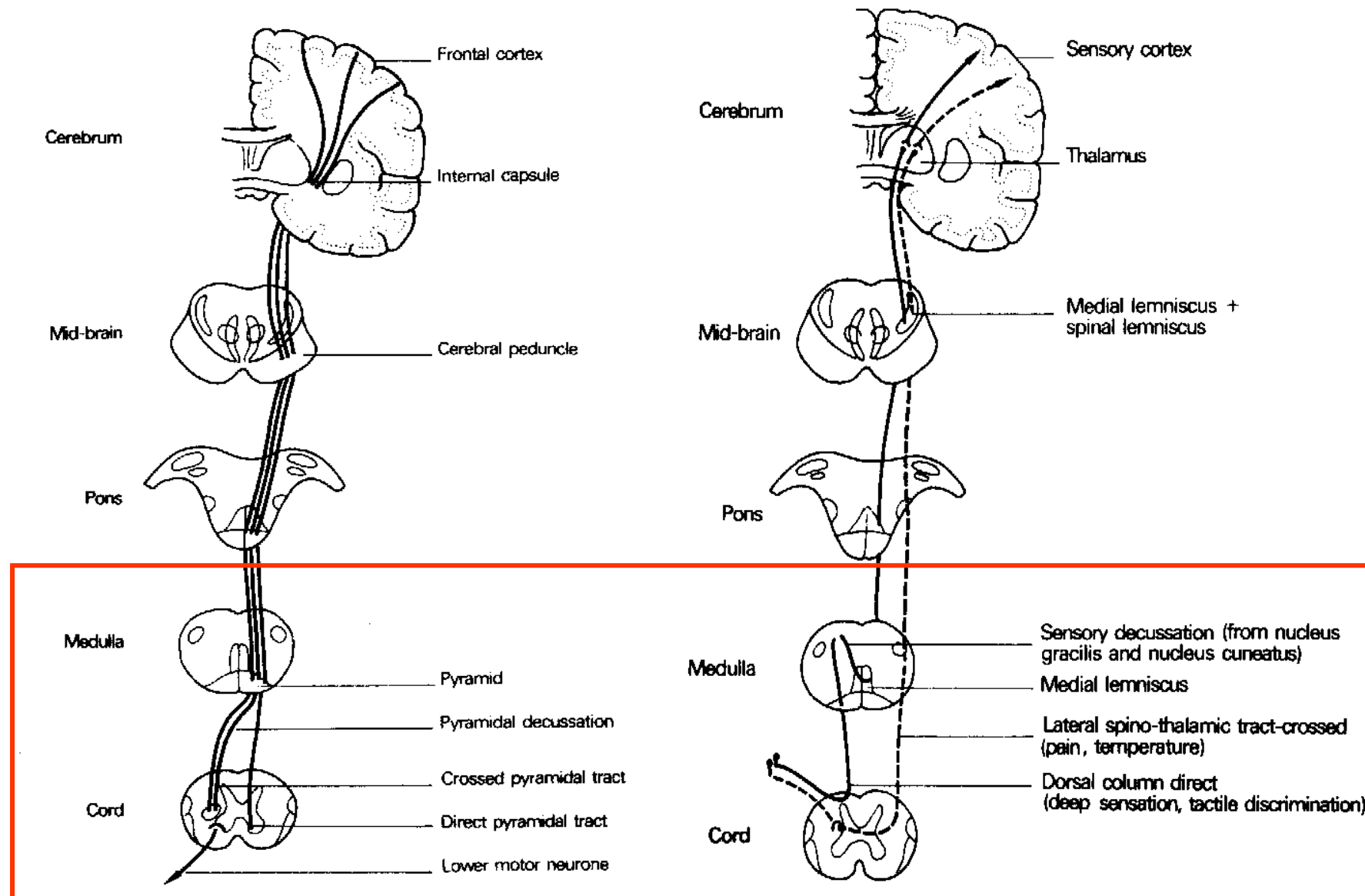


Incomplete injury / spinal cord syndromes



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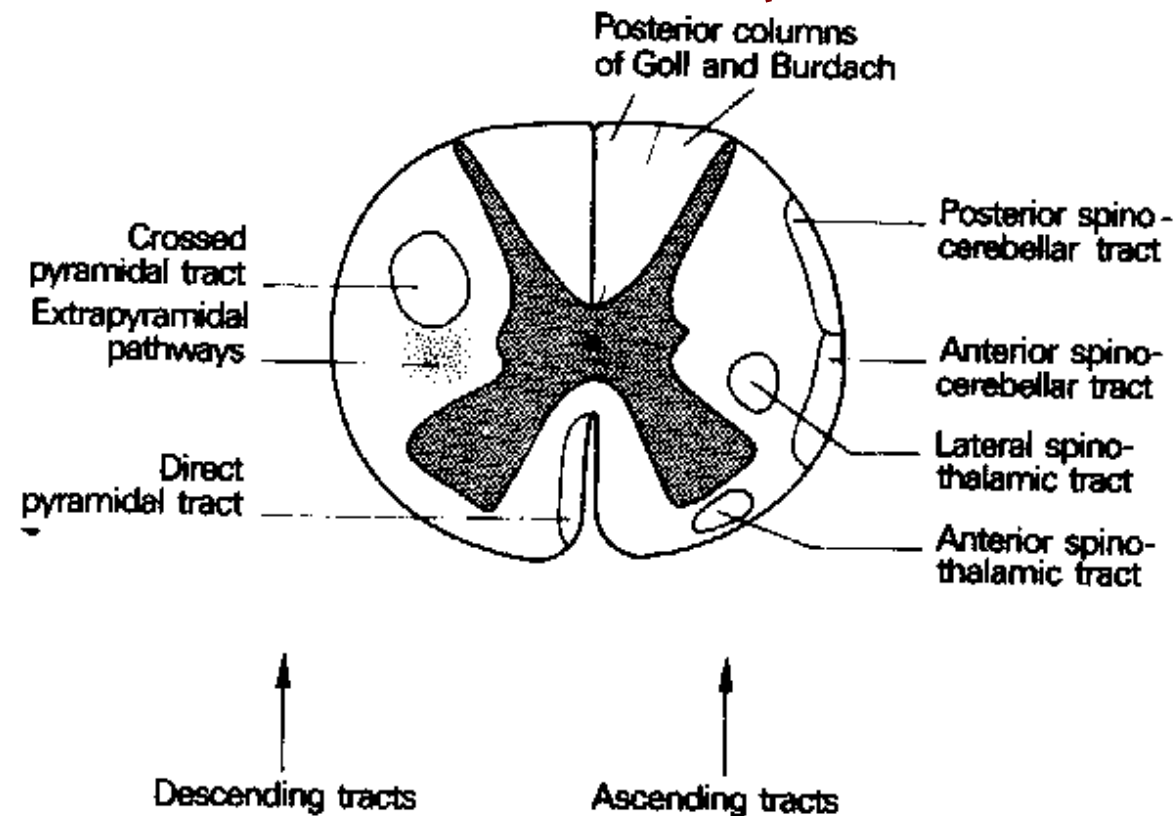
- Central Cord Syndrome
- Brown-Sequard Syndrome
- Anterior Cord Syndrome
- Posterior Cord Syndrome
- Mixed
- Cauda Equina Syndrome



Motor tracts

Sensory tracts

Fine touch & proprioception -
uncrossed



Motor - *crossed*

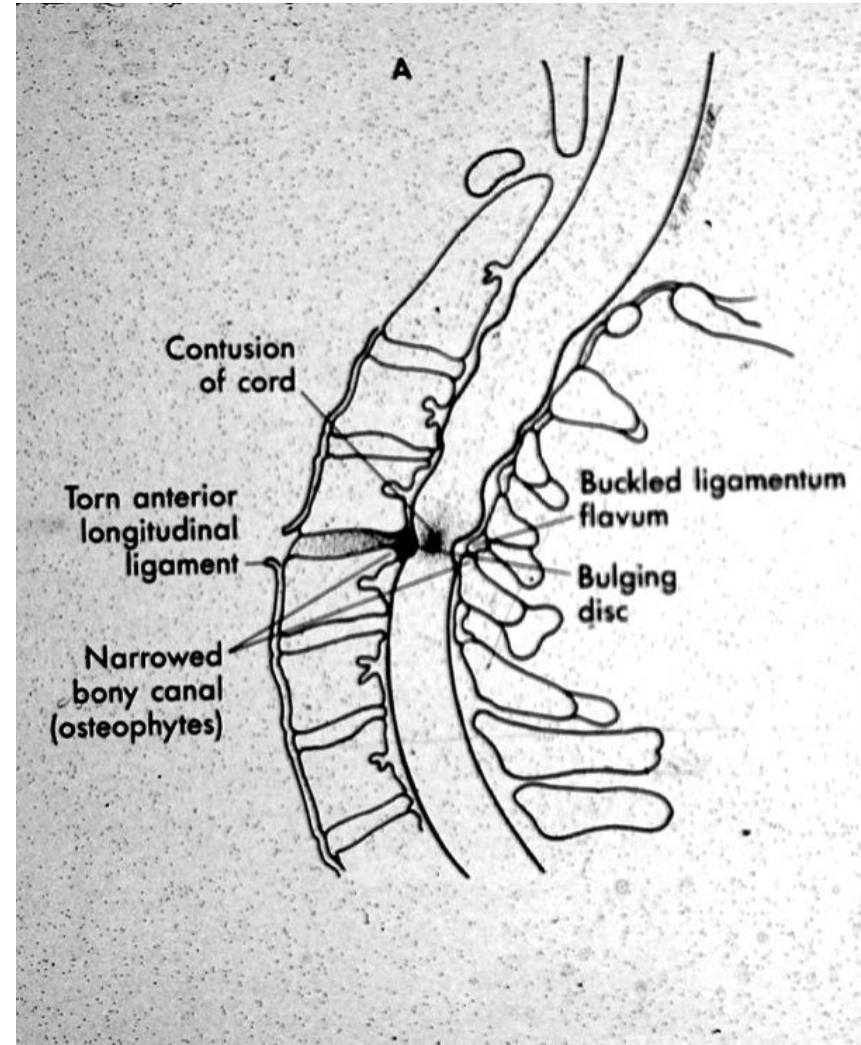
Pain and temperature
crossed

Central Cord Syndrome



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- Most common
- Hyperextension injury in degenerative cervical spine

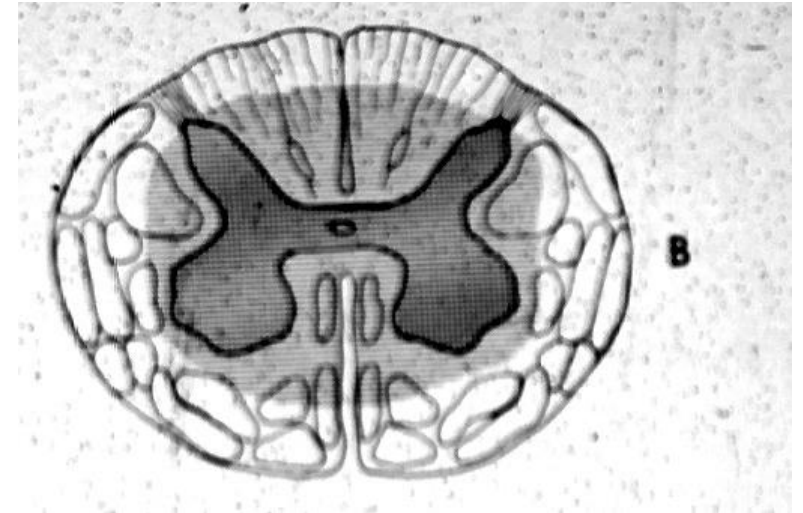


Central Cord Syndrome



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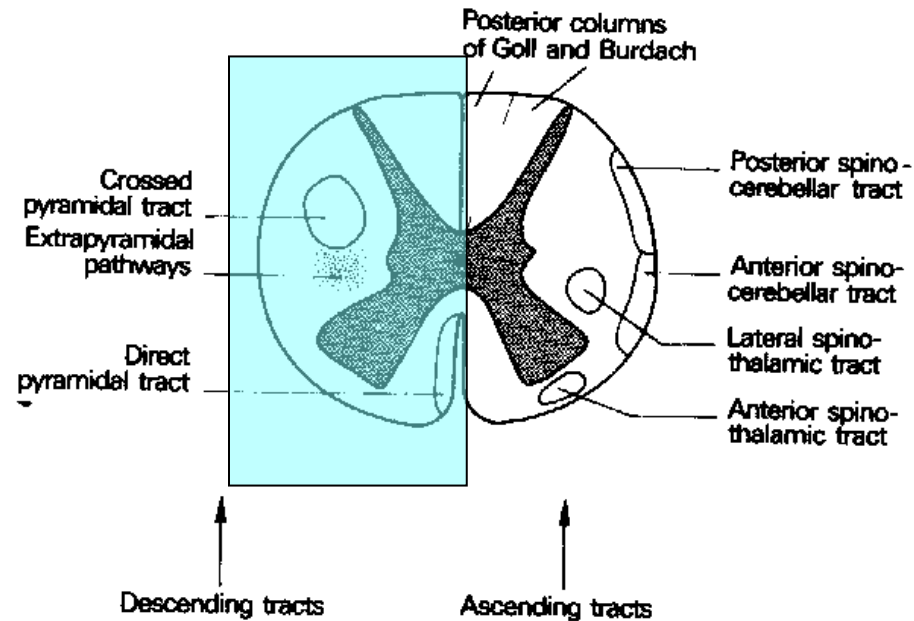
- Quadriparesis initially, with recovery
- Arm affected more than legs
- Sensory sparing variable
- Most patients regain ambulation
- Persistent distal upper limb neurological deficits common



Brown-Sequard Syndrome



- Motor weakness on side of injury
- Contralateral loss of pain and temperature
- Unilateral fracture / subluxation
- Prognosis good

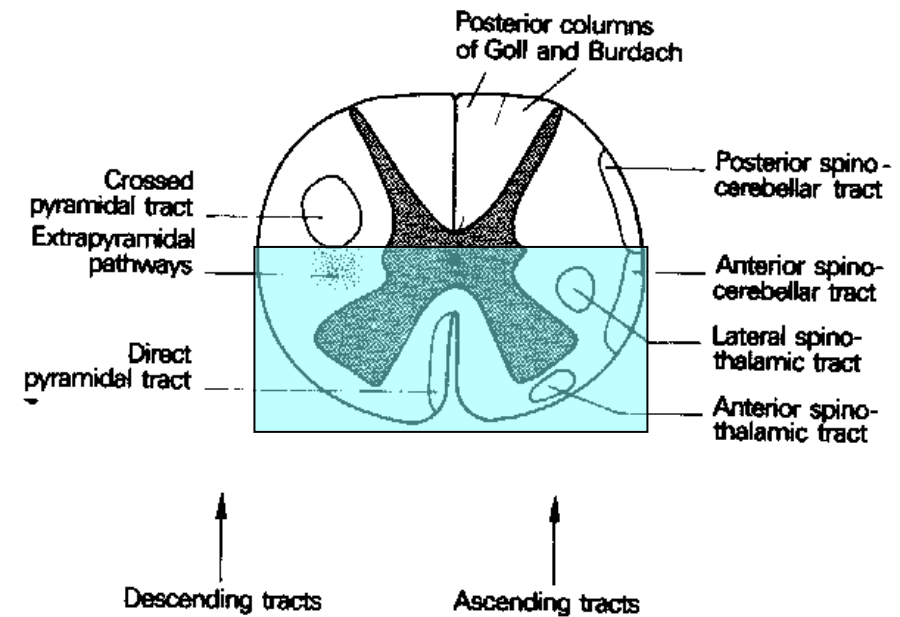


Anterior Cord Syndrome



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- Hyperflexion injury with bone or disc fragment
- Anterior spinal artery compression
- Motor and pain/temperature loss
- Posterior column preserved
- Prognosis poor



- Primary and secondary injury
 - Maximal damage occurs at the time of injury (primary injury)
 - Inappropriate or delayed management can result in further oedema and cord damage (secondary injury) which is preventable

Minimizing secondary injury



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- Proper transfer
- *Steroid*
- Decompress
- Stabilise

Steroids and SCI



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- National Acute Spinal Cord Injury Study (NASCIS)
- Methylprednisolone given within 8 hours of injury had better motor and sensory function at 6 weeks and 6 months
- Higher rates of wound infection, pneumonia, sepsis so overall benefit is questionable

Indications for Surgery

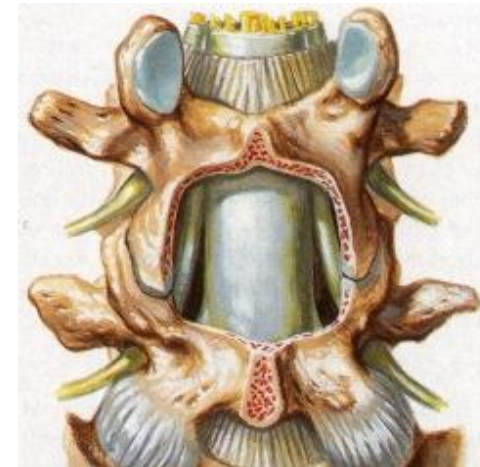


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- Structural
 - Instability
- Decompression
 - Neurological deficit
 - Lack of improvement / deteriorating neurology
- Polytrauma
 - To facilitate mobilization and rehabilitation

Choice of surgery

- Decompression
 - Anterior
 - Posterior
- Stabilisation
 - Internal
 - Anterior
 - Posterior
 - *External*
 - Non-operative management, maximizes immobilization



Complications following SCI



Table 1
Organ system complications following spinal cord injury

Organ System	Complications
Cardiovascular	Bradycardia/dysrhythmia Cardiac arrest Cardiogenic pulmonary edema
Pulmonary	Hypoventilation/respiratory failure Poor secretion control Acute respiratory distress syndrome Aspiration Pneumonia
Gastrointestinal	Gastric dysmotility Adynamic ileus Gastritis and ulceration Pancreatitis
Hematologic	Venothromboembolism
Neurologic	Neurogenic shock Depression Posttraumatic stress disorder Anxiety Autonomic dysreflexia
Genitourinary	Bladder dysfunction Urinary tract infection Priapism
Integument	Pressure ulceration

Rehabilitation



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- Multiple team approach
- Develop the functions which remain normal
- Compensate for those that have been lost

Respiratory problems



- Respiratory muscle paralysis
 - Abdominal / intercostals: T1 – T12
 - Diaphragm: C3 - C5
 - Accessory muscles (trapezius, sternomastoid): above C3
- Decreased cough effort
- Direct chest injuries
 - haemopneumothorax
 - pulmonary contusion
- Autonomic dysfunction

Treatment of respiratory problems



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- Vigorous PT and postural drainage
- Suction
- Tracheostomy
- Bronchoscopy

DVT and pressure sores



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- Deep vein thrombosis
 - Mechanical prophylaxis via sequential compression device
 - Anticoagulation prophylaxis within 72 hours of injury
- Pressure sores
 - Turning every two hours
 - Padding of at-risk areas
 - Frequent skin inspection

Functional goals by level of spinal cord injury



TABLE 39-3

Functional Abilities by Level of Spinal Cord Injury

LEVEL	FUNCTION	ADL	ELIMINATION	MOBILITY
C ₁ –C ₃	No movement or sensation below the neck; Ventilator-dependent	D	D	Voice or sip-n-puff controlled electric WC
C ₄	Movement and sensation of head and neck; some partial function of the diaphragm	D	D	Chin-operated electric WC
C ₅	Controls head, neck, and shoulders; flexes elbows	WA	D	Electric WC
C ₆	Uses shoulder, extends wrist	I or WA	WA	WC, self-transfer
C ₇ and C ₈	Extends elbow, flexes wrist, some use of fingers	I	I	Manual WC
T ₁ –T ₅	Has full hand and finger control, full use of thoracic muscles	I	I	Manual WC
T ₆ –T ₁₀	Controls abdominal muscles, has good balance	I	I	Manual WC
T ₁₁ –L ₅	Flexes and abducts the hips; flexes and extends the knees	I	I	Ambulates with leg braces, short braces, or canes
S ₁ –S ₅	Full control of legs; progressive bowel, bladder, and sexual function	I	I	

D = dependent, I = independent, WA = with assistance, WC = wheelchair.



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